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Biologically Degradable MULTI-COMPONENT POLYMER FIBRES

Abstract

The invention relates to a biologically degradable multi-component polymer fibre, in particular bi-component fibres, with advantageous physical properties, to a process for its production, as well as to its use.

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Background/Summary

[0001] The invention relates to a biologically degradable polymer fibre with advantageous physical properties, to a process for its production, as well as to its use.

[0002] Polymer fibres, i.e. fibres based on synthetic polymers, are produced industrially on a huge scale. In this regard, the basic synthetic polymer is processed using a melt spinning process. To this end, the thermoplastic polymeric material is melted and fed in the liquid state into a spinning beam by means of an extruder. From this spinning beam, the molten material is fed to what are known as spinnerets. The spinneret usually comprises a spinneret plate provided with a plurality of holes out of which the individual capillaries (filaments) of the fibre are extruded. In addition to the melt spinning process, wet spinning or solvent spinning processes are also used for the production of spun fibres. Here, instead of a melt, a highly viscous solution of a synthetic polymer is extruded through dies with fine holes. When large numbers of individual spinnerets are charged at the same time and in parallel by the same flow of polymer melt, the person skilled in the art terms this a multiple spinning process.

[0003] The polymer fibres produced in this manner are used for textile and/or technical applications. In these applications, it is of advantage for the polymer fibres to have high mechanical strength so that the post-processing of the fibres can be carried out without problems, for example by drawing on rolling mills. It is also advantageous for the polymer fibres, in particular when in the form of nonwovens, to have low thermal shrinkage.

[0004] Modifying or equipping polymer fibres for the respective end use or for the necessary intermediate treatment steps, for example drawing and/or crimping, is usually carried out by applying suitable softening agents or dressings which are applied to the surface of the prepared polymer fibres or the polymer fibres to be treated.

[0005] A further possibility for modification is the chemical modification of the polymer skeleton itself, for example by incorporating flame-retardant co-monomers into the polymeric main and/or side chain.

[0006] Furthermore, additives, for example antistatic agents or coloured pigments, can be introduced into the molten thermoplastic polymer or can be introduced into the polymer fibre during the multiple spinning process.

[0007] Recently, there has been a surge in the development of fibre systems which on the one hand satisfy the requirements set out above, and furthermore which exhibit good biological degradability, and on the other hand require few or no changes, so that existing processes and equipment can still be used.

[0008] Recently, there has been an additional surge in the development of fibre systems which on the one hand satisfy the requirements set out above, and furthermore which preferably can be produced at least in part from sustainable raw materials, and on the other hand require few or no changes, so that existing processes and equipment can still be used.

[0009] In biologically degradable fibres, because of the chronological sequence of the degradation processes, which are influenced by various factors, there is often a poor and poorly controllable relationship between maximum service life of the product and the time period over which the anticipated biological degradation takes place.

[0010] Thus, there is a need for the provision of a polymer fibre the biological degradability of which can be tailored to the intended end use and also which is compatible with existing post-processing of fibres.

[0011] The present invention allows the degradation behaviour of the fibre to be controlled by using two components which behave differently with respect to each other as regards degradation.

[0012] The said need is satisfied by means of the multi-component polymer fibre in accordance with the invention, wherein the polymer fibre: [0013] (i) comprises at least one component A and at least one component B, [0014] (ii) the component A comprises a thermoplastic polymer A, [0015] (iii) the component B comprises a thermoplastic polymer B, characterized in that [0016] (iv) the component A additionally has at least one additive A which increases the biological degradability of the multi-component fibre and the component B does not have an additive B which increases the biological degradability of the multi-component fibre, [0017] or [0018] (v) the component B additionally has at least one additive B which increases the biological degradability of the multi-component fibre and the component A does not have an additive A which increases the biological degradability of the multi-component fibre, [0019] or [0020] (vi) the component A additionally has at least one additive A and the component B additionally has at least one additive B which together increase the biological degradability of the multi-component fibre, with the proviso that when (i) the thermoplastic polymer A and the thermoplastic polymer B are identical, the additives A and B are different, or (ii) when the additives A and B are identical, thermoplastic polymer A and thermoplastic polymer B are different.

[0021] In the context of the present invention, an increased biological degradability of the multi-component fibre means that this multi-component fibre, compared with a multi-component fibre without additives A and/or B, is degraded more rapidly, wherein the determination is carried out in accordance with at least one method selected from the group (i) ASTM D5338-15 (2021) (Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials Under Controlled Composting Conditions, Incorporating Thermophilic Temperatures (DOI: 10.1520/D5338-15R21) ASTM International, West Conshohocken, PA, 2015, www.astm.org), (ii) ASTM D6400-12 (Standard Specification for Labeling of Plastics Designed to be Aerobically Composted in Municipal or Industrial Facilities) (DOI: 10.1520/D6400-12), (iii) ASTM D5511 (ASTM D5511-11 Standard Test Method for Determining Anaerobic Biodegradation of Plastic Materials Under High-Solids Anaerobic Digestion Conditions (DOI: 10.1520/D5511-11) and ASTM D5511-18 Standard Test Method for Determining Anaerobic Biodegradation of Plastic Materials Under High-Solids Anaerobic-Digestion Conditions; (DOI: 10.1520/D5511-18)), (iv) ASTM D6691 (ASTM D6691-09 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in the Marine Environment by a Defined Microbial Consortium or Natural Sea Water Inoculum) (DOI: 10.1520/D6691-09) and ASTM D6691-17, Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in the Marine Environment by a Defined Microbial Consortium or Natural Sea Water Inoculum (DOI: 10.1520/D6691-17)), (v) ASTM D5210-92 (Anaerobic Degradation in the Presence of Sewage Sludge) (DOI: 10.1520/D5210-92), (vi) PAS 9017:2020 (Plastics-Biodegradation of polyolefins in an open-air terrestrial environment-Specification), ISBN 978 0 539 17478 6; 2021 Oct. 31, (vii) ASTM D5988 (ASTM D5988-12 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in Soil) (DOI: 10.1520/D5988-12), ASTM D5988-18 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in Soil (DOI: 10.1520/D5988-18), ASTM D5988-03 Standard Test Method for Determining Aerobic Biodegradation in Soil of Plastic Materials or Residual Plastic Materials After Composting (DOI: 10.1520/D5988-03)), (viii) EN 13432:2000-12 Packaging-Requirements for packaging recoverable through composting and biodegradation-Test scheme and evaluation criteria for the final acceptance of packaging; German version EN 13432:2000 (DOI: 10.31030/9010637), (ix) ISO 14855-1:2013-04 (DOI: 10.31030/1939267) and ISO 14855-2:2018-07 (ICS 83.080.01) Determination of the ultimate aerobic biodegradability of plastic materials under controlled

composting conditions (Method by analysis of evolved carbon dioxide), (x) EN 14995:2007-03-Plastics-Evaluation of compostability (DOI: 10.31030/9730527) or (xi) ISO 17088:2021-04 (Specifications for compostable plastics) (ICS 83.080.01).

[0022] When processed using the (staple fibre) spinning process, the multi-component polymer fibre in accordance with the invention is usually deposited as a tow and subsequently drawn on a rolling mill using the usual methods and then post-treated. The tow can also be processed further directly and so deposition of the tow in what are known as cans can be entirely or partially dispensed with.

[0023] When processed using the (filament) spinning process, the multi-component polymer fibre in accordance with the invention can be cooled directly following exit from the spinneret and drawn and deposited on a collecting belt or wound onto bobbins. The filaments may be drawn further for further processing in order to increase the orientation of the molecular chains, in particular with a draw ratio between 0.5 and 3. Furthermore, it is possible to texturize the filaments.

[0024] The combination of different biological degradabilities for the components A and B means that the biological degradability of the products resulting from these multi-component polymer fibres can be designed and customized.

[0025] Textile fabrics, for example nonwovens, can be produced from the multi-component polymer fibres in accordance with the invention. When the textile fabrics, in particular nonwovens, are consolidated using thermobonding, it is advantageous for the melting point of the thermoplastic polymer in component A to be at least 5° C. higher than the melting point of the thermoplastic polymer in component B. In this embodiment, the multi-component polymer fibres are preferably bi-component fibres in which component A forms the core and component B forms the shell. Particularly preferably, the melting point of the thermoplastic polymer in component A is at least 10° C. higher than the melting point of the thermoplastic polymer in component B.

[0026] During thermobonding, the fibres are bonded together at the contact or crossover points. When the component B formed from thermoplastic polymer B with additive B has a higher biological degradability than component A formed from thermoplastic polymer A with additive A, the contact or crossover points of the fibres are degraded together first and the textile fabric, for example a nonwoven, disintegrates faster, whereupon the overall degradability is increased.

[0027] Furthermore, it is possible to provide a multi-component fibre which comprises a very rapidly biologically degradable component A with at least one further component B, wherein component B has a lower biological degradation rate than component A. In this manner, a staged biological degradation of the fibres can be obtained, giving rise to technical advantages, for example a warning of mechanical failure, a comparatively high residual stability of the fibres with advanced biological degradation, etc.

[0028] Further possible arrangements of the components in the multi-component fibre in addition to the core/shell structure, wherein the core may be concentric with and also may be eccentric with respect to the shell, are a side-by-side structure, a matrix-fibril structure as well as slice-of-cake structures or orange-slice structures.

[0029] Furthermore, it is possible to provide multi-component polymer fibres, in particular bi-component polymer fibres, which combine a very rapidly biologically degradable core (component A) produced from thermoplastic polymer A and optionally an additive A, with an equally biologically degradable shell (component B) produced from thermoplastic polymer B with additive B, so that the component A is only biologically degraded when component B has already been biologically degraded. This is intended to accelerate the degradation, which sets in as soon as component B has been degraded to a sufficient extent.

[0030] Thus, in a further aspect, the present invention provides a bi-component fibre with a core/shell structure wherein, [0031] (i) the component A forms the core and the component B forms the shell of the fibre, [0032] (ii) the component A in the core comprises thermoplastic polymer A, [0033] (iii) the component B comprises a thermoplastic polymer B, [0034] (iv) the melting point of

the thermoplastic polymer in the component A in the core is at least 5° C. higher than the melting point of the thermoplastic polymer in the component B in the shell; preferably, the melting point is at least 10° C. higher, characterized in that [0035] (v) the component A has a higher biological degradability than the component B; preferably, the component A has at least one additive A, [0036] or [0037] (vi) the component B has a higher biological degradability than the component A; [0038] preferably, the component B has at least one additive B.

[0039] The higher biological degradability is determined in accordance with at least one method selected from the group formed by: [0040] (i) ASTM D5338-15 (2021) Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials Under Controlled Composting Conditions, Incorporating Thermophilic Temperatures (DOI: 10.1520/D5338-15R21) ASTM International, West Conshohocken, PA, 2015, www.astm.org), [0041] (ii) ASTM D6400-12 (Standard Specification for Labeling of Plastics Designed to be Aerobically Composted in Municipal or Industrial Facilities) (DOI: 10.1520/D6400-12), [0042] (iii) ASTM D5511 (ASTM D5511-11 Standard Test Method for Determining Anaerobic Biodegradation of Plastic Materials Under High-Solids Anaerobic-Digestion Conditions (DOI: 10.1520/D5511-11) and ASTM D5511-18 Standard Test Method for Determining Anaerobic Biodegradation of Plastic Materials Under High-Solids Anaerobic-Digestion Conditions; (DOI: 10.1520/D5511-18)), [0043] (iv) ASTM D6691 (ASTM D6691-09 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in the Marine Environment by a Defined Microbial Consortium or Natural Sea Water Inoculum) (DOI: 10.1520/D6691-09) and ASTM D6691-17, Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in the Marine Environment by a Defined Microbial Consortium or Natural Sea Water Inoculum (DOI: 10.1520/D6691-17)), [0044] (v) ASTM D5210-92 (Anaerobic Degradation in the Presence of Sewage Sludge) (DOI: 10.1520/D5210-92), [0045] (vi) PAS 9017:2020 (Plastics-Biodegradation of polyolefins in an open-air terrestrial environment-Specification), ISBN 978 0 539 17478 6; 2021 Oct. 31, [0046] (vii) ASTM D5988 (ASTM D5988-12 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in Soil) (DOI: 10.1520/D5988-12), ASTM D5988-18 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in Soil (DOI: 10.1520/D5988-18), ASTM D5988-03 Standard Test Method for Determining Aerobic Biodegradation in Soil of Plastic Materials or Residual Plastic Materials After Composting (DOI: 10.1520/D5988-03)), [0047] (viii) EN 13432:2000-12 Packaging-Requirements for packaging recoverable through composting and biodegradation-Test scheme and evaluation criteria for the final acceptance of packaging; German version EN 13432:2000 (DOI: 10.31030/9010637), [0048] (ix) ISO 14855-1:2013-04 (DOI: 10.31030/1939267) and ISO 14855-2:2018-07 (ICS 83.080.01) Determination of the ultimate aerobic biodegradability of plastic materials under controlled composting conditions (Method by analysis of evolved carbon dioxide), [0049] (x) EN 14995:2007-03-Plastics-Evaluation of compostability (DOI: 10.31030/9730527) or [0050] (xi) ISO 17088:2021-04 (Specifications for compostable plastics) (ICS 83.080.01).

[0051] The bi-component fibre in accordance with the invention can therefore be tailored for any intended purpose and any environment.

[0052] Because the component A has a higher biological degradability than component B, firstly, the shell component B—which protects against biological degradability—is biologically degraded and after it has degraded, component A is degraded. In this way, materials may be used as component A which have a biological degradability which is so high that they usually cannot be engineered, because their high biological degradability means that they are assumed to be unstable or unsuitable. The protective shell can also have a retarding action, i.e. the shell initially at least slows down the biological degradability and after a specific time or period of use, a rapid biological degradation occurs.

[0053] Thus, for example, textile fabrics which have the one bi-component fibre in accordance with the invention, may be used in agriculture, in which the component A has a high biological

degradability in accordance with ASTM D5338-15 or ASTM D6400 or ASTM D5988, but is initially protected by the shell. Textile fabrics of this type can be disposed of after the intended use by means of controlled composting.

[0054] A further advantage of the present invention is that textile fabrics which have a bi-component fibre in accordance with the invention can be provided which on the one hand, for example in agriculture, can be used as intended, but which can reach the ocean via rivers in the event of incorrect disposal. For this purpose, it is advantageous to use a component A which has a high biological degradability in accordance with ASTM D6691. Because incorrect disposal usually breaks or damages the protective shell, a controlled biological degradability, for example in a maritime environment, is ensured.

[0055] Because component B has a higher biological degradability than component A, initially the shell component B is degraded, which leads to a faster disintegration of textile fabrics which have the bi-component fibres in accordance with the invention. In this manner, for example, following their intended use, hygiene articles can be composted in a controlled manner in household waste or a sewage plant.

[0056] In this manner, a step-wise biological degradation may be obtained, bringing with it technical advantages, for example signalling mechanical failure, a comparatively high residual stability of the fibres in the case of advanced biological degradation, etc.

[0057] The bi-component fibre in accordance with the invention may be a finite-length fibre, for example what is known as a staple fibre, or a continuous fibre (filament). There are no critical restrictions to the length of the aforementioned staple fibres, but in general they are 2 to 200 mm, preferably 3 to 120 mm, particularly preferably 4 to 60 mm.

[0058] The individual linear density of the bi-component fibre in accordance with the invention, preferably staple fibres, is preferably between 0.5 and 30 dtex, in particular 0.7 to 13 dtex. For some applications, linear densities of between 0.5 and 3 dtex and fibre lengths of <10 mm, in particular <8 mm, particularly preferably <6 mm, particularly preferably <5 mm, are particularly suitable.

[0059] The cross-sectional proportion of the core with respect to the total cross sectional area of the fibres is between 20% and 90% and the cross-sectional proportion of the shell with respect to the total cross sectional area of the fibres is between 80% and 10%.

[0060] The ratio of the cross sectional area of component A and component B may also contribute to fine tuning the biological degradability behaviour of the fibre.

[0061] Particularly preferred bi-component polymer fibres are those in which the additive A and/or additive B are selected from the group (i) basic alkali and/or alkaline earth compounds (pH>7 dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly preferably CaCO_3 , and alkaline additives, particularly preferably CaO, (ii) aliphatic polyesters, (iii) sugars, in particular mono-saccharides, di-saccharides and oligo-saccharides, (iv) catalysts for transesterifications, in particular under basic conditions, (v) carbohydrates, in particular starch and/or cellulose, as well as mixtures thereof.

[0062] Particularly preferred bi-component polymer fibres are those in which the thermoplastic polymer A and/or the thermoplastic polymer B comprises at least one polyester and the additive A and/or additive B is selected from the group (i) basic alkali and/or alkaline earth compounds (pH>7 dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly preferably CaCO_3 , and alkaline additives, particularly preferably CaO, (ii) aliphatic polyesters, (iii) sugars, in particular mono-saccharides, di-saccharides and oligo-saccharides, (iv) catalysts for transesterifications, in particular under basic conditions, (v) carbohydrates, in particular starch and/or cellulose, as well as mixtures thereof. The aforementioned aliphatic polyesters are distinguished from the polyesters of the thermoplastic polymer A and polymer B in respect of their chemical nature, i.e. the polyester of the thermoplastic polymer A and polymer B is an aliphatic polyester or copolyester, which has been produced from polyols and aliphatic and/or

aromatic dicarboxylic acids or their derivatives (anhydrides, esters) by means of polycondensation. Particularly preferred additives A and/or additives B contain at least two substances, wherein preferred combinations are: [0063] A) basic alkali and/or alkaline earth compounds (pH>7 dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly preferably CaCO_3 , and alkaline additives, particularly preferably CaO in combination with catalysts for transesterifications, in particular under basic conditions; [0064] B) sugars, in particular mono-saccharides, di-saccharides and oligo-saccharides, in combination with carbohydrates, in particular starch and/or cellulose, as well as mixtures thereof; [0065] C) aliphatic polyesters, optionally in combination with sugars, in particular mono-saccharides, di-saccharides and oligo-saccharides, or carbohydrates, in particular starch and/or cellulose, as well as mixtures thereof. [0066] Most preferred additives A for partially aromatic “araliphatic” polyester or copolyester as thermoplastic polymer A are containing at least [0067] basic alkali and/or alkaline earth compounds (pH>7 dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly preferably CaCO_3 , and alkaline additives, particularly preferably CaO, preferably in combination with catalysts for transesterifications, in particular under basic conditions; and [0068] aliphatic polyesters, especially aliphatic polyester having no side chain carbon atoms, optionally in combination with (i) sugars, in particular mono-saccharides, di-saccharides and oligo-saccharides, (ii) carbohydrates, in particular starch and/or (iii) cellulose, as well as mixtures thereof.

[0069] Out of the aforementioned particularly preferred bi-component polymer fibres are those preferred in which the thermoplastic polymer A is polyester and the thermoplastic polymer B is a polyester being different from the polyester in polymer A, and preferably is a co-polyester, and—each—the additive A and the additive B is independently selected from the combination of [0070] basic alkali and/or alkaline earth compounds (pH>7 dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly preferably CaCO_3 , and alkaline additives, particularly preferably CaO, preferably in combination with catalysts for transesterifications, in particular under basic conditions; and [0071] aliphatic polyesters, especially aliphatic polyester having no side chain carbon atoms, optionally in combination with (i) sugars, in particular mono-saccharides, di-saccharides and oligo-saccharides, (ii) carbohydrates, in particular starch and/or (iii) cellulose, as well as mixtures thereof.

[0072] Particularly preferred bi-component polymer fibres are those in which the thermoplastic polymer B is a polyolefin, in particular a polypropylene polymer, which includes as additive B at least (i) metal compounds, in particular transition metal compounds, as well as their salts, preferably at least two chemically different transition metal compounds and (ii) unsaturated carboxylic acids or their anhydrides/esters/amides, preferably in combination with synthetic rubber and/or natural rubber, and—optionally—further comprising (iii) sugars, in particular monosaccharides, disaccharides and oligosaccharides, (iv) carbohydrates, in particular starch and/or (v) cellulose, as well as mixtures thereof. Further, phenolic antioxidant stabilizer and CaO can be present.

[0073] The biological degradability can be fine tuned by means of the quantity of additive A in component A or additive B in component B. The quantity of additive is usually between 0.005% by weight and 20% by weight, particularly preferably between 0.01% by weight and 5% by weight, with respect to the total quantity of component A or component B.

[0074] Among the additives described above, the following in particular are suitable: (i) basic alkali and/or alkaline earth compounds (pH>7 dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly preferably CaCO_3 , (ii) sugars, in particular mono-saccharides, di-saccharides and oligo-saccharides, as well as (iii) carbohydrates, in particular starch and/or cellulose, as well as mixtures thereof, as well as the aforementioned combinations A), B) or C), because their degradability in accordance with ASTM D6691 or in accordance with ASTM

D5338-15, ASTM D6400 or ASTM D5988 can specifically be adjusted.

Thermoplastic Polymers

[0075] The polymers used in accordance with the invention are thermoplastic polymers.

[0076] The term “thermoplastic polymer” as used in the present invention means a synthetic material which can be deformed in a specific range of temperatures, preferably in the range 25° C. to 350° C., (thermoplastic). This procedure is reversible, i.e. it can be put into its viscous state any number of times by cooling and heating again, as long as the material is not damaged too much by overheating, which causes what is known as thermal decomposition, or by shaping the material under mechanical load. This is the difference between thermoplastic polymers and thermosets and elastomers.

[0077] The thermoplastic polymers used in accordance with the invention are preferably polymers selected from the group formed by acrylonitrile-ethylene-propylene-(diene)-styrene copolymer, acrylonitrile-methacrylate copolymer, acrylonitrile-methylmethacrylate copolymer, chlorinated acrylonitrile, polyethylene-styrene copolymer, acrylonitrile-butadiene-styrene copolymer, acrylonitrile-ethylene-propylene-styrene copolymer, cellulose acetobutyrate, cellulose acetopropionate, hydrated cellulose, carboxymethylcellulose, cellulose nitrate, cellulose propionate, cellulose triacetate, polyvinyl chloride, ethylene-acrylic acid copolymer, ethylene-butylacrylate copolymer, ethylene-chlorotrifluoroethylene copolymer, ethylene-ethylacrylate copolymer, ethylene-methacrylate copolymer, ethylene-methacrylic acid copolymer, ethylene-tetrafluoroethylene copolymer, ethylene-vinyl alcohol copolymer, ethylene-butene copolymer, ethylcellulose, polystyrene, polyfluoroethylene-propylene, methylmethacrylate-acrylonitrile-butadiene styrene copolymer, methylmethacrylate-butadiene-styrene copolymer, methylcellulose, polyamide 11, polyamide 12, polyamide 46, polyamide 6, polyamide 6-3-T, polyamide 6-terephthalic acid copolymer, polyamide 66, polyamide 69, polyamide 610, polyamide 612, polyamide 61, polyamide MXD 6, polyamide PDA-T, polyamide, polyarylether, polyaryletherketone, polyamideimide, polyarylamide, polyamino-bis-maleimide, polyarylate, polybutene-1, polybutylacrylate, polybenzimidazole, poly-bis-maleimide, polyoxadiazobenzimidazole, polybutylterephthalate, polycarbonate, polychlorotrifluoroethylene, polyethylene, polyestercarbonate, polyaryletherketone, polyetheretherketone, polyetherimide, polyetherketone, polyethylene oxide, polyarylether sulphone, polyethylene terephthalate, polyimide, polyisobutylene, polyisocyanurate, polyimide sulphone, polymethacrylimide, polymethacrylate, poly-4-methylpentene, polyacetal, polypropylene, polyphenyl oxide, polypropylene oxide, polyphenylene sulphide, polyphenylene sulphone, polystyrene, polysulphone, polytetrafluoroethylene, polyurethane, polyvinyl acetate, polyvinyl alcohol, polyvinylbutyral, polyvinyl chloride, polyvinylidene chloride, polyvinylidene fluoride, polyvinyl fluoride, polyvinyl methyl ether, polyvinylpyrrolidone, styrene-butadiene copolymer, styrene-isoprene copolymer, styrene-maleic acid anhydride copolymer, styrene-maleic acid anhydride-butadiene copolymer, styrene methyl methacrylate copolymer, styrene-methyl styrene copolymer, styrene-acrylonitrile copolymer, vinyl chloride-ethylene copolymer, vinyl chloride-methacrylate copolymer, vinyl chloride-maleic acid anhydride copolymer, vinyl chloride-maleimide copolymer, vinyl chloride-methylmethacrylate copolymer, vinyl chloride-octyl acrylate copolymer, vinyl chloride-vinyl acetate copolymer, vinyl chloride-vinylidene chloride copolymer, vinyl chloride-vinylidene chloride-acrylonitrile copolymer.

[0078] Among the thermoplastic polymers, melt spinnable synthetic biopolymers are preferred, particularly preferably polycondensates and polymerisates produced from bio-based starting materials.

[0079] The term “synthetic biopolymer” as used in the present invention designates a substance which primarily consists of biogenic raw materials (sustainable raw materials). This differentiates them from conventional mineral oil-based substances or plastics such as, for example, polyethylene (PE), polypropylene (PP) and polyvinyl chloride (PVC), as long as their feedstock is not renewable

(e.g. bio-PE/green PE).

[0080] In a preferred embodiment, the multi-component fibres in accordance with the invention are produced from biologically degradable synthetic biopolymers, wherein the term “biologically degradable” may, for example, be specified, tested and/or determined in accordance with at least one method selected from the group formed by (i) ASTM D5338-15 (2021) (Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials Under Controlled Composting Conditions, Incorporating Thermophilic Temperatures (DOI: 10.1520/D5338-15R21) ASTM International, West Conshohocken, PA, 2015, www.astm.org), (ii) ASTM D6400-12 (Standard Specification for Labelling of Plastics Designed to be Aerobically Composted in Municipal or Industrial Facilities) (DOI: 10.1520/D6400-12), (iii) ASTM D5511 (ASTM D5511-11 Standard Test Method for Determining Anaerobic Biodegradation of Plastic Materials Under High-Solids Anaerobic digestion Conditions (DOI: 10.1520/D5511-11) and ASTM D5511-18 Standard Test Method for Determining Anaerobic Biodegradation of Plastic Materials Under High-Solids Anaerobic-digestion Conditions; (DOI: 10.1520/D5511-18)), (iv) ASTM D6691 (ASTM D6691-09 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in the Marine Environment by a Defined Microbial Consortium or Natural Sea Water Inoculum) (DOI: 10.1520/D6691-09) and ASTM D6691-17, Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in the Marine Environment by a Defined Microbial Consortium or Natural Sea Water Inoculum (DOI: 10.1520/D6691-17)), (v) ASTM D5210-92 (Anaerobic Degradation in the Presence of Sewage Sludge) (DOI: 10.1520/D5210-92), (vi) PAS 9017:2020 (Plastics—Biodegradation of polyolefins in an open-air terrestrial environment—Specification), ISBN 978 0 539 17478 6; 2021 Oct. 31, (vii) ASTM D5988 (ASTM D5988-12 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in Soil) (DOI: 10.1520/D5988-12), ASTM D5988-18 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in Soil (DOI: 10.1520/D5988-18), ASTM D5988-03 Standard Test Method for Determining Aerobic Biodegradation in Soil of Plastic Materials or Residual Plastic Materials After Composting (DOI: 10.1520/D5988-03)), (viii) EN 13432:2000-12 Packaging-Requirements for packaging recoverable through composting and biodegradation-Test scheme and evaluation criteria for the final acceptance of packaging; German version EN 13432:2000 (DOI: 10.31030/9010637), (ix) ISO 14855-1:2013-04 (DOI: 10.31030/1939267) and ISO 14855-2:2018-07 (ICS 83.080.01) Determination of the ultimate aerobic biodegradability of plastic materials under controlled composting conditions (Method by analysis of evolved carbon dioxide), (x) EN 14995:2007-03-Plastics-Evaluation of compostability (DOI: 10.31030/9730527) or (xi) ISO 17088:2021-04 (Specifications for compostable plastics) (ICS 83.080.01).

[0081] Preferred synthetic biopolymers in the context of the present invention are aliphatic, araliphatic polyesters or copolyesters which are produced from polyols, and aliphatic and/or aromatic dicarboxylic acids or their derivatives (anhydrides, esters) by polycondensation, wherein the polyols may be substituted or unsubstituted, and the polyols may be linear or branched polyols.

[0082] Preferred polyols are polyols containing 2 to 8 carbon atoms, polyalkylene etherglycols containing 2 to 8 carbon atoms and cycloaliphatic diols containing 4 to 12 carbon atoms. Non-limiting examples of polyols which may be used include ethylene glycol, diethylene glycol, propylene glycol, 1,3-propanediol, 2,2-dimethyl-1,3-propanediol, 2-methyl-1,3-propanediol, 1,3-butanediol, 1,4-butanediol, 1,5-pentanediol, 1,6-hexanediol, polyethylene glycol, diethylene glycol, 2,2,4-trimethyl-1,6-hexanediol, thiodiethanol, 1,3-cyclohexanedimethanol, 1,4-cyclohexanedimethanol, 2,2,4,4-tetramethyl-1,3-cyclobutanediol, triethylene glycol and tetraethylene glycol. Preferred polyols include 1,4-butanediol, 1,3-propanediol, ethylene glycol, 1,6-hexanediol, diethylene glycol, isosorbitol and 1,4-cyclohexanedimethanol.

[0083] Preferred aliphatic dicarboxylic acids include substituted or unsubstituted, linear or branched, non-aromatic dicarboxylic acids selected from the group formed by aliphatic dicarboxylic acids containing 2 to 12 carbon atoms and cycloaliphatic dicarboxylic acids

containing 5 to 10 carbon atoms, wherein the cycloaliphatic dicarboxylic acids may also contain heteroatoms in the ring.

[0084] The substituted non-aromatic dicarboxylic acids typically contain 1 to 4 substituents selected from halogens, C6-C10 aryl and C1-C4 alkoxy. Non-limiting examples of aliphatic and cycloaliphatic dicarboxylic acids include maleic acid, succinic acid, glutaric acid, adipic acid, pimelic acid, azelaic acid, sebacic acid, fumaric acid, 2,2-dimethylglutaric acid, suberic acid, 1,3-cyclopentane dicarboxylic acid, 1,4-cyclohexane dicarboxylic acid, 3-cyclohexanedicarboxylic acid, diglycolic acid, itaconic acid, maleic acid, 2,5-norbornane dicarboxylic acid.

[0085] Preferred aromatic dicarboxylic acids include substituted or unsubstituted aromatic dicarboxylic acids selected from the group formed by aromatic dicarboxylic acids containing 6 to 12 carbon atoms, wherein these carboxylic acids may also comprise heteroatoms in the aromatic ring and/or in the substituents.

[0086] The substituted aromatic dicarboxylic acids may typically contain 1 to 4 substituents selected from halogens, C6-C10 aryl and C1-C4 alkoxy. Non-limiting examples of aromatic dicarboxylic acids include phthalic acid, isophthalic acid, terephthalic acid, naphthalene dicarboxylic acid and furan dicarboxylic acid.

[0087] The aforementioned aliphatic dicarboxylic acids may also be together with the aforementioned aromatic dicarboxylic acids in the form of copolymers or terpolymers; non-limiting examples are polybutylene-adipate-terephthalate and bio-based PTA.

[0088] Particularly preferred synthetic biopolymers in the context of the present invention are aliphatic polyesters with repeat units of at least 4 carbon atoms, for example polyhydroxyalkanoates such as polyhydroxyvalerate and polyhydroxybutyrate-hydroxyvalerate copolymer, polycaprolactone, furan dicarboxylic acid, and succinate-based aliphatic polymers (for example polybutylene succinate, polybutylene succinate adipate and polyethylene succinate). Special examples may be selected from polyethylene oxalate, polyethylene malonate, polyethylene succinate, polypropylene oxalate, polypropylene malonate, polypropylene succinate, polybutylene oxalate, polybutylene malonate, polybutylene succinate and blends and copolymers of these compounds.

[0089] In particular, the preferred synthetic biopolymers are aliphatic polyesters comprising repeat units of lactic acid (PLA), hydroxy fatty acid (PHF) (also designated as polyhydroxyalkanoate, PHA), in particular hydroxybutanoic acid (PHB) and succinate-based aliphatic polymers, for example polybutylene succinate, polybutylene succinate adipate and polyethylene succinate.

[0090] The "aliphatic polyesters" should be understood to mean those polyesters which typically have at least approximately 50% molar, preferably at least approximately 60% molar, particularly preferably at least approximately 70% molar, particularly preferably at least 95% molar aliphatic monomers.

[0091] In the context of the present invention, moreover, thermoplastic polymers with a glass transition temperature of more than -125°C ., advantageously more than -30°C ., preferably more than 30°C ., particularly preferably more than 50°C ., in particular more than 70°C ., are extremely advantageous. In the context of a more particularly preferred embodiment of the present invention, the glass transition temperature of the polymer is in the range -125°C . to 200°C ., in particular in the range -125°C . to 100°C .

[0092] Among the thermoplastic synthetic biopolymers, the glass transition temperature is preferably more than 20°C ., advantageously more than 25°C ., preferably more than 30°C ., particularly preferably more than 35°C ., in particular more than 40°C . In the context of a more particularly preferred embodiment of the present invention, the glass transition temperature of the polymer is in the range 35°C . to 55°C ., in particular in the range 40°C . to 50°C .

[0093] Particularly preferred polyesters are PET with a glass transition temperature of at least 70°C ., PLA with a glass transition temperature in the range 40°C . to 70°C ., PHA and PHB with a glass transition temperature in the range -40°C . to 62°C ., PBS, as well as PBS copolymers such

as PBSA with a glass transition temperature in the range -45°C. to 45°C. and polycaprolactone with a glass transition temperature in the range -75°C. to 45°C.

[0094] Polyesters, in particular polyethylene terephthalate, usually have a molecular weight corresponding to an intrinsic viscosity (IV) of 0.4 to 1.4 (dl/g), measured for solutions in dichloroacetic acid at 25°C.

[0095] Particularly preferred polyesters are those such as PET, PEN, PLA, PBS, PEIT with a number average molecular weight (M_n), preferably determined by gel permeation chromatography against polystyrene standards with a narrow distribution or by end group titration, of at least 20000 g/mol. Better still, the polydispersibility of these polymers is at least 1.7.

[0096] Polyesters of particular interest are those such as PET with a melting point between 250°C. and 260°C.

[0097] Particularly interesting polyesters are those such as PET with a melting enthalpy of (80%: 43 J/g; 100% crystal/theoretical): 115 J/g.

[0098] Polyesters of particular interest are those such as PET with a crystallization temperature of at least 125°C. and a crystallization enthalpy (125°C.) of at least 31 J/g.

[0099] Polyesters of particular interest are those which are commercially available from Trevira GmbH, for example such as Trevira® T298.

[0100] Particularly preferred polyamides have a glass transition temperature in the range 30°C. to 80°C. , in particular in the range 35°C. to 65°C. , particularly preferably in the range 50°C. to 60°C. , wherein these values are intended for PA 6.6 and PA 6 in particular.

[0101] Polyamides of particular interest are those such as PA 6.6 and PA 6 with a number average molecular weight (M_n), preferably determined by gel permeation chromatography against polystyrene standards with a narrow distribution or by end group titration, of at least 10000 g/mol.

[0102] Polyamides of particular interest are those such as PA 6.6 and PA 6, with a melting point between 170°C. and 280°C. , more preferably between 200°C. and 260°C. Polyamides of particular interest are those such as PA 6.6 and PA 6 with a Crystal melting enthalpy (100% crystal) of 190°C.

[0103] Particularly interesting polyamides are those such as PA 6.6 and PA 6 with a softening temperature of 204°C.

[0104] Commercially available polyamides such as Nylon, Perlon or Grilon are of particular interest.

[0105] Polyolefins of particular interest are those such as polyethylene (PE) or polypropylene (PP) homopolymers, as well as copolymers or terpolymers which comprise at least 50 mol % of ethylene and/or propylene repeat units.

[0106] Polyethylenes of particular interest are low density polyethylene (LDPE), linear low density polyethylene (LLDPE), very low density polyethylene (VLDPE), ultra low density polyethylene (ULDPE), medium density polyethylene (MDPE), polymethylpentene (PMP), polybutene-1 (PB-1); ethylene-octene copolymers, stereoblock PP, olefin block copolymers, propylene-butane copolymers.

[0107] Particularly preferred polyolefins are PE with a glass transition temperature in the range -100°C. to -35°C. and PP with a glass transition temperature in the range -10°C. to -5°C.

[0108] Polyethylenes of particular interest are those with a melting point between 120°C. and 135°C. and polypropylene with a melting point between 158°C. and 170°C.

[0109] Polyethylenes of particular interest are those with a crystal melting enthalpy (100% crystal) of 290 J/g and polypropylene with a crystal melting enthalpy of 190 J/g.

[0110] Commercially available polyolefins such as LDPE (PE Aspun 6834, Dow), HDPE (SKGC MK 910), PP (Braskem) such as Braskem HSP165G, are of particular interest.

[0111] Further suitable polymers are those which have a melting temperature of more than 50°C. , advantageously at least 75°C. , preferably more than 150°C. Particularly preferably, the melting temperature is in the range 120°C. to 285°C. , in particular in the range 150°C. to 270°C. ,

particularly preferably in the range 175° C. to 270° C.

[0112] In this regard, the glass transition temperature and the melting temperature of the polymer are preferably determined by means of Differential Scanning Calorimetry (DSC).

[0113] Particularly preferred synthetic biopolymers in accordance with the invention are thermoplastic polycondensates based on what are known as biopolymers, which contain the repeat units of lactic acid, hydroxybutyric acid, succinic acid, glycolic acid and/or furan dicarboxylic acid, preferably lactic acid and/or glycolic acid, in particular lactic acid. Polylactic acids are particularly preferred in this regard.

[0114] A variety of high melting point synthetic biopolymers (melting point between 110° C. and 270° C., preferably between 140° C. and 270° C., more preferably between 180° C. and 270° C.), such as polyesters, may be used in the present invention, such as polyesteramides, modified polyethylene terephthalate, polylactic acid (PLA), terpolymers based on polylactic acid, polybutylene succinate, polyalkylene furanoate such as PEF, polyglycolic acid, polyalkylene carbonates (such as polyethylene carbonate), polyhydroxyalkanoates (PHA) such as polyhydroxybutyrates (PHB), polyhydroxyvalerates (PHV) or polyhydroxybutyrate-hydroxyvalerate copolymers (PHBV).

[0115] The term “polylactic acid” (PLA) should be understood here to mean polymers which are constructed from lactic acid units. Such polylactic acids are usually produced by condensation of lactic acids, but are also obtained by ring-opening polymerization of lactides under suitable conditions.

[0116] Particularly suitable polylactic acids in accordance with the invention include poly(glycolide-co-L-lactide), poly(L-lactide), poly(L-lactide-co-ε-caprolactone), poly(L-lactide-co-glycolide), poly(L-lactide-co-D,L-lactide), poly(D,L-lactide-co-glycolide) as well as poly(dioxanone). As an example, polymers of this type are commercially available from Boehringer Ingelheim Pharma KG (Germany) under the trade names Resomer® GL 903, Resomer® L 206 S, Resomer® L 207 S, Resomer® L 209 S, Resomer® L 210, Resomer® L 210 S, Resomer® LC 703 S, Resomer® LG 824 S, Resomer® LG 855 S, Resomer® LG 857 S, Resomer® LR 704 S, Resomer® LR 706 S, Resomer® LR 708, Resomer® LR 927 S, Resomer® RG 509 S and Resomer® X 206 S from Biomer, Inc. (Germany) with the name Biomer™ L9000. Other suitable polylactic acid polymers are commercially available from Natureworks, LLC, Minneapolis, Minnesota, USA.

[0117] Especially advantageous polylactic acids for the purposes of the present invention are poly-D-, poly-L- or poly-D, L-lactic acids in particular.

[0118] The expression “polylactic acid” generally refers to homopolymers of lactic acid such as poly(L-lactic acid), poly(D-lactic acid), poly(DL-lactic acid), mixtures thereof and copolymers, which contain lactic acid as the primary component and a small proportion, preferably less than 10% molar, of a co-polymerizable co-monomer.

[0119] Further suitable materials are copolymers or terpolymers based on polylactic acid, polyglycolic acid, polyalkylene carbonates (such as polyethylene carbonate), polyhydroxyalkanoates (PHA), polyhydroxybutyrates (PHB), polyhydroxyvalerates (PHV) and polyhydroxybutyrate-hydroxyvalerate copolymers (PHBV).

[0120] In a particularly preferred embodiment, the biopolymer is exclusively a thermoplastic polycondensate based on lactic acids.

[0121] The polylactic acids used in accordance with the invention preferably have a number average molecular weight (M_n) which is a minimum of 500 g/mol, preferably a minimum of 1000 g/mol, particularly preferably a minimum of 5000 g/mol, appropriately a minimum of 10000 g/mol, in particular a minimum of 25000 g/mol. On the other hand, the number average is preferably a maximum of 1000000 g/mol, appropriately a maximum of 500000 g/mol, advantageously a maximum of 100000 g/mol, in particular a maximum of 50000 g/mol. A number average molecular weight in the range from a minimum of 10000 g/mol to 500000 g/mol has proved to be particularly

advantageous in the context of the present invention.

[0122] The mass average molecular weight (Mw) of preferred lactic acid polymers, in particular of poly-D-, poly-L- or poly-D, L-lactic acids, is preferably in the range 750 g/mol to 5000000 g/mol, preferably in the range 5000 g/mol to 1000000 g/mol, particularly preferably in the range 10000 g/mol to 500000 g/mol, in particular in the range 30000 g/mol to 500000 g/mol, and the polydispersity of these polymers is advantageously in the range 1.5 to 5.

[0123] The inherent viscosity of particularly suitable lactic acid polymers, poly-D-, poly-L- or poly-D,L-lactic acids in particular, measured in chloroform at 25° C., 0.1% polymer concentration, is in the range 0.5 dl/g to 8.0 dl/g, preferably in the range 0.8 dl/g to 7.0 dl/g, in particular in the range 1.5 dl/g to 3.2 dl/g.

[0124] Furthermore, the inherent viscosity of particularly suitable lactic acid polymers, in particular poly-D-, poly-L- or poly-D,L-lactic acids, measured in hexafluoro-2-propanol at 30° C., at 0.1% polymer concentration, is in the range from 1.0 dl/g to 2.6 dl/g, in particular in the range from 1.3 dl/g to 2.3 dl/g.

[0125] Of particular interest are polylactic acids with a glass transition temperature between 50° C. and 65° C.

[0126] Of particular interest are polylactic acids with a melting point between 155° C. and 180° C.

[0127] Of particular interest are commercially available polylactic acids such as NatureWorks PLA 6202D.

[0128] The term “polyhydroxy fatty acid esters” (PHF) as used in the context of the invention should preferably be understood to mean the following polymers: poly(3-hydroxypropionate) (PHP), poly(3-hydroxybutyrate) (PHB, P3HB), poly(3-hydroxyvalerate) (PHV), poly(3-hydroxyhexanoate) (PHHx), poly(3-hydroxyheptanoate) (PHH), poly(3-hydroxyoctanoate) (PHO), poly(3-hydroxynonanoate) (PHN), poly(3-hydroxydecanoate) (PHD), poly(3-hydroxyundecanoate) (PHUD), poly(3-hydroxydodecanoate) (PHDD), poly(3-hydroxytetradecanoate) (PHTD), poly(3-hydroxypentadecanoate) (PHPD), poly(3-hydroxyhexadecanoate) (PHHxD) as well as blends of the aforementioned polymers.

[0129] In addition to the aforementioned homopolymers, polyhydroxy fatty acid ester copolymers such as poly(3-hydroxypropionate-co-3-hydroxybutyrate) (P3HP-3HB), poly(3-hydroxypropionate-co-4-hydroxybutyrate) (P3HP-4HB), poly(3-hydroxybutyrate-co-4-hydroxybutyrate) (P (3HB-4HB)), poly(3-hydroxybutyrate-co-3-hydroxyvalerate) (PHBV), poly(3-hydroxybutyrate-co-3-hydroxyvalerate-co-3-hydroxyhexanoate) (PHBV-HHx) as well as blends of the aforementioned copolymers, may be used together or with the aforementioned homopolymers.

[0130] The thermoplastic polyhydroxy fatty acid ester polymers used in accordance with the invention are commercially available; examples are Mirel, Biomer P 209, Biopol, Aonilex X, Proganic.

[0131] The thermoplastic polyhydroxy fatty acid ester polymers used in accordance with the invention preferably have a glass transition temperature in the range -2° C. to 62° C.

[0132] The thermoplastic polyhydroxy fatty acid ester polymers used in accordance with the invention preferably have a melting temperature in the range 100° C. to 177° C.

[0133] The thermoplastic polyhydroxy fatty acid ester polymers used in accordance with the invention preferably have a melt flow index (MFI) of 5-10 g/10 min (190° C., 2.16 kg) determined in accordance with ISO 1133-1:2011.

[0134] The thermoplastic polyhydroxy fatty acid ester polymers used in accordance with the invention preferably have a number average molecular weight (Mn) of at least 200000 Dalton, in particular at least 220000 Dalton, particularly preferably at least 250000 Dalton, and a maximum of up to 3000000 Dalton, in particular up to 2500000 Dalton, particularly preferably up to 2000000 Dalton.

[0135] The thermoplastic polyhydroxy fatty acid ester polymers used in accordance with the invention usually have a mass average molecular weight (Mw) which is about a factor of 2,

preferably a factor of 3, times the number average molecular weight (Mn).

[0136] The term “succinate-based aliphatic polymers” should be understood to mean polymers with the following general formula:

##STR00001##

wherein R.sub.1, R.sub.2, R.sub.3, R.sub.4 represent linear or branched aliphatic hydrocarbon residues consisting of 2 to 20 carbon atoms.

[0137] Examples in this regard are polybutylene succinate, polybutylene succinate adipate and polyethylene succinate.

[0138] The thermoplastic succinate-based aliphatic polymers used in accordance with the invention are commercially available; examples are Bionolle 1000, BioPBS.

[0139] The thermoplastic succinate polymers used in accordance with the invention preferably have a glass transition temperature in the range -45°C. to 45°C.

[0140] The thermoplastic succinate polymers used in accordance with the invention preferably have a crystallization temperature in the range 70°C. to 90°C.

[0141] The thermoplastic succinate polymers used in accordance with the invention preferably have a melting temperature in the range 60°C. to 180°C.

[0142] The thermoplastic succinate polymers used in accordance with the invention preferably have a melt flow index (MFI) of 5-10 g/10 min (190°C. , 2.16 kg), determined in accordance with ISO 1133-1:2011.

[0143] The thermoplastic succinate polymers used in accordance with the invention preferably have a number average molecular weight (Mn) of at least 20000 Dalton, in particular at least 30000 Dalton, particularly preferably at least 35000 Dalton, and a maximum of up to 140000 Dalton, in particular up to 120000 Dalton, particularly preferably up to 110000 Dalton.

[0144] The thermoplastic succinate polymers used in accordance with the invention preferably have a mass average molecular weight (Mw) which is about a factor of 2, preferably a factor of 3, times the number average molecular weight (Mn).

[0145] Polycaprolactone (PCL) is a synthetic biopolymer within the meaning of the present invention.

[0146] Of particular interest are polycaprolactones with a glass transition temperature between -45°C. and 45°C.

[0147] Of particular interest are polycaprolactones with a crystallization temperature between 70°C. and 90°C.

[0148] Of particular interest are polycaprolactones with a melting point between 60°C. and 180°C.

[0149] Of particular interest are polycaprolactones with a melting enthalpy of 70-145 J/g.

[0150] Of particular interest are polycaprolactones with a number average molecular weight (Mn), preferably determined by gel permeation chromatography against polystyrene standards with a narrow distribution or by end group titration, of at least 20000 Dalton to 140000 Dalton.

[0151] Of particular interest are commercially available polycaprolactones such as Resomer C 209.

Thermoplastic Polymer A

[0152] The thermoplastic polymer A is selected from the aforementioned group of thermoplastic polymers.

[0153] Among the thermoplastic polymers A, melt spinnable synthetic biopolymers are preferred, particularly preferably polycondensates and polymerisates produced from bio-based starting materials. The synthetic biopolymer is selected from the aforementioned group of synthetic biopolymers.

[0154] Preferred synthetic biopolymers are aliphatic, araliphatic polyesters or copolyesters which are produced from polyols, and aliphatic and/or aromatic dicarboxylic acids or their derivatives (anhydrides, esters) by polycondensation, wherein the polyols may be substituted or unsubstituted, linear or branched polyols.

[0155] Preferred polyols are polyols containing 2 to 8 carbon atoms, polyalkylene etherglycols containing 2 to 8 carbon atoms and cycloaliphatic diols containing 4 to 12 carbon atoms. Non-limiting examples of polyols which may be used are ethylene glycol, diethylene glycol, propylene glycol, 1,3-propanediol, 2,2-dimethyl-1,3-propanediol, 2-methyl-1,3-propanediol, 1,3-butanediol, 1,4-butanediol, 1,5-pentanediol, 1,6-hexanediol, polyethylene glycol, diethylene glycol, 2,2,4-trimethyl-1,6-hexanediol, thiodiethanol, 1,3-cyclohexanedimethanol, 1,4-cyclohexanedimethanol, 2,2,4,4-tetramethyl-1,3-cyclobutanediol, triethylene glycol and tetraethylene glycol. Preferred polyols include 1,4-butanediol, 1,3-propanediol, ethylene glycol, 1,6-hexanediol, diethylene glycol, isosorbitol and 1,4-cyclohexanedimethanol.

[0156] Preferred aliphatic dicarboxylic acids include substituted or unsubstituted, linear or branched, non-aromatic dicarboxylic acids selected from the group formed by aliphatic dicarboxylic acids containing 2 to 12 carbon atoms and cycloaliphatic dicarboxylic acids containing 5 to 10 carbon atoms, wherein the cycloaliphatic dicarboxylic acids may also contain heteroatoms in the ring.

[0157] The substituted non-aromatic dicarboxylic acids typically contain 1 to 4 substituents selected from halogens, C6-C10 aryl and C1-C4 alkoxy. Non-limiting examples of aliphatic and cycloaliphatic dicarboxylic acids include maleic acid, succinic acid, glutaric acid, adipic acid, pimelic acid, azelaic acid, sebacic acid, fumaric acid, 2,2-dimethylglutaric acid, suberic acid, 1,3-cyclopentane dicarboxylic acid, 1,4-cyclohexanedicarboxylic acid, 3-cyclohexanedicarboxylic acid, diglycolic acid, itaconic acid, maleic acid, 2,5-norbornane dicarboxylic acid.

[0158] Preferred aromatic dicarboxylic acids include substituted or unsubstituted, aromatic dicarboxylic acids selected from the group formed by aromatic dicarboxylic acids containing 6 to 12 carbon atoms, wherein these carboxylic acids may also comprise heteroatoms in the aromatic ring and/or in the substituents.

[0159] The substituted aromatic dicarboxylic acids may typically have 1 to 4 substituents selected from halogens, C6-C10 aryl and C1-C4 alkoxy. Non-limiting examples of aromatic dicarboxylic acids include phthalic acid, isophthalic acid, terephthalic acid, naphthalene dicarboxylic acid and furan dicarboxylic acid.

[0160] Together with the aforementioned aromatic dicarboxylic acids, the aforementioned aliphatic dicarboxylic acids may also be present in the form of copolymers or terpolymers; non-limiting examples are polybutylene-adipate terephthalate and bio-based PTA, for example.

[0161] Among the thermoplastic polymers A, preferred melt spinnable synthetic biopolymers are aliphatic polyesters with repeat units of at least 4 carbon atoms, for example polyhydroxyalkanoates such as polyhydroxyvalerate and polyhydroxybutyrate-hydroxyvalerate copolymer, polycaprolactone, furan dicarboxylic acid, and succinate-based aliphatic polymers (for example polybutylene succinate, polybutylene succinate adipate and polyethylene succinate). Special examples may be selected from polyethylene oxalate, polyethylene malonate, polyethylene succinate, polypropylene oxalate, polypropylene malonate, polypropylene succinate, polybutylene oxalate, polybutylene malonate, polybutylene succinate and blends and copolymers of these compounds.

[0162] Particularly preferred synthetic biopolymers are aliphatic polyesters comprising repeat units of lactic acid (PLA), hydroxy fatty acid (PHF) (also known as polyhydroxyalkanoate, PHA), in particular hydroxybutanoic acid (PHB) and succinate-based aliphatic polymers, for example polybutylene succinate, polybutylene succinate adipate and polyethylene succinate.

[0163] "Aliphatic polyesters" should be understood to mean those polyesters which typically have at least approximately 50% molar, preferably at least approximately 60% molar, particularly preferably at least approximately 70% molar, particularly preferably at least 95% molar aliphatic monomers.

[0164] Among the thermoplastic polymers A, thermoplastic polymers with a glass transition temperature of more than -125°C ., advantageously more than -30°C ., preferably more than 30°C

C., particularly preferably more than 50° C., in particular more than 70° C., are preferred. In the context of a particularly preferred embodiment, the glass transition temperature of the polymer is in the range -125° C. to 200° C., in particular in the range -125° C. to 100° C.

[0165] Among the thermoplastic polymers A, thermoplastic synthetic biopolymers which are preferred are those with a glass transition temperature which is preferably more than 20° C., advantageously more than 25° C., preferably more than 30° C., particularly preferably more than 35° C., in particular more than 40° C. In the context of a particularly preferred embodiment, the glass transition temperature of the polymer is in the range 35° C. to 55° C., in particular in the range 40° C. to 50° C.

[0166] Particularly preferred polyesters are PET with a glass transition temperature of at least 70° C., PLA with a glass transition temperature in the range 40° C. to 70° C., PHA and PHB with a glass transition temperature in the range -40° C. to 62° C., PBS as well as PBS copolymers such as PBSA with a glass transition temperature in the range -45° C. to 45° C. and polycaprolactone with a glass transition temperature in the range -75° C. to 45° C.

[0167] Polyesters, in particular polyethylene terephthalate, usually have a molecular weight corresponding to an intrinsic viscosity (IV) of 0.4 to 1.4 (dl/g), measured for solutions in dichloroacetic acid at 25° C.

[0168] Polyesters of particular interest are those such as PET, PEN, PLA, PBS, PEIT with a number average molecular weight (Mn), preferably determined by gel permeation chromatography against polystyrene standards with a narrow distribution or by end group titration, of at least 20000 g/mol. Better still, the polydispersibility of these polymers is at least 1.7.

[0169] Polyesters of particular interest are those such as PET with a melting point between 250° C. and 260° C.

[0170] Particularly interesting polyesters are those such as PET with a melting enthalpy of (80%: 43 J/g; 100% crystal/theoretical): 115 J/g.

[0171] Polyesters of particular interest are those such as PET with a crystallization temperature of at least 125° C. and a crystallization enthalpy (125° C.) of at least 31 J/g.

[0172] Polyesters of particular interest are those which are commercially available from Trevira GmbH, for example such as Trevira® T298.

[0173] Particularly preferred polyamides have a glass transition temperature in the range 30° C. to 80° C., in particular in the range 35° C. to 65° C., particularly preferably in the range 50° C. to 60° C., wherein these values are intended for PA 6.6 and PA 6 in particular.

[0174] Polyamides of particular interest are those such as PA 6.6 and PA 6 with a number average molecular weight (Mn), preferably determined by gel permeation chromatography against polystyrene standards with a narrow distribution or by end group titration, of at least 10000 g/mol.

[0175] Polyamides of particular interest are those such as PA 6.6 and PA 6, with a melting point between 170° C. and 280° C., more preferably between 200° C. and 260° C. Polyamides of particular interest are those such as PA 6.6 and PA 6 with a crystallization melting enthalpy (100% crystal) of 190° C.

[0176] Particularly interesting polyamides are those such as PA 6.6 and PA 6 with a softening temperature of 204° C.

[0177] Commercially available polyamides such as Nylon, Perlon or Grilon are of particular interest.

[0178] Polyolefins of particular interest are those such as polyethylene (PE) or polypropylene (PP) homopolymers, as well as copolymers or terpolymers which comprise at least 50 mol % of ethylene and/or propylene repeat units.

[0179] Polyethylenes of particular interest are low density polyethylene (LDPE), linear low density polyethylene (LLDPE), very low density polyethylene (VLDPE), ultra low density polyethylene (ULDPE), medium density polyethylene (MDPE), polymethylpentene (PMP), polybutene-1 (PB-1); ethylene-octene copolymers, stereoblock PP, olefin block copolymers, propylene-butane

copolymers.

[0180] Particularly preferred polyolefins are PE with a glass transition temperature in the range -100°C. to -35°C. and PP with a glass transition temperature in the range -10°C. to -5°C.

[0181] Polyethylenes of particular interest are those with a melting point between 120°C. and 135°C. and polypropylene with a melting point between 158°C. and 170°C.

[0182] Polyethylenes of particular interest are those with a crystallization melting enthalpy (100% crystal) of 290 J/g and polypropylene with a crystallization melting enthalpy of 190 J/g .

[0183] Of particular interest are commercially available polyolefins such as LDPE (PE Aspun 6834, Dow), HDPE (SKGC MK 910), PP (Braskem).

[0184] Further suitable polymers are those which have a melting temperature of more than 50°C. , advantageously at least 75°C. , preferably of more than 150°C. Particularly preferably, the melting temperature is in the range from 120°C. to 285°C. , in particular in the range from 150°C. to 270°C. , particularly preferably in the range from 175°C. to 270°C.

[0185] In this regard, the glass transition temperature and the melting temperature of the polymer are preferably determined by means of Differential Scanning Calorimetry (DSC).

[0186] Particularly preferred synthetic biopolymers in accordance with the invention are thermoplastic polycondensates based on what are known as biopolymers, which contain the repeat units of lactic acid, hydroxybutyric acid, succinic acid, glycolic acid and/or furan dicarboxylic acid, preferably lactic acid and/or glycolic acid, in particular lactic acid. Polylactic acids are particularly preferred in this regard.

[0187] A variety of high melting point synthetic biopolymers (melting point between 110°C. and 270°C. , preferably between 140°C. and 270°C. , more preferably between 180°C. and 270°C.), such as polyesters, may be used in the present invention, such as polyesteramides, modified polyethylene terephthalate, polylactic acid (PLA), terpolymers based on polylactic acid, polybutylene succinate, polyalkylene furanoate such as PEF, polyglycolic acid, polyalkylene carbonates (such as polyethylene carbonate), polyhydroxyalkanoates (PHA) such as polyhydroxybutyrate (PHB), polyhydroxyvalerate (PHV) or polyhydroxybutyrate-hydroxyvalerate copolymers (PHBV).

[0188] The term “polylactic acid” (PLA) should be understood to mean polymers which are constructed from lactic acid units. Such polylactic acids are usually produced by condensation of lactic acids, but are also obtained by ring-opening polymerization of lactides under suitable conditions.

[0189] Particularly suitable polylactic acids in accordance with the invention include poly(glycolide-co-L-lactide), poly(L-lactide), poly(L-lactide-co-s-caprolactone), poly(L-lactide-co-glycolide), poly(L-lactide-co-D,L-lactide), poly(D,L-lactide-co-glycolide) as well as poly(dioxanone). As an example, polymers of this type are commercially available from Boehringer Ingelheim Pharma KG (Germany) under the trade names Resomer® GL 903, Resomer® L 206 S, Resomer® L 207 S, Resomer® L 209 S, Resomer® L 210, Resomer® L 210 S, Resomer® LC 703 S, Resomer® LG 824 S, Resomer® LG 855 S, Resomer® LG 857 S, Resomer® LR 704 S, Resomer® LR 706 S, Resomer® LR 708, Resomer® LR 927 S, Resomer® RG 509 S and Resomer® X 206 S, from Biomer, Inc. (Germany) with the name Biomer™ L9000. Other suitable polylactic acid polymers are commercially available from Natureworks, LLC, Minneapolis, Minnesota, USA.

[0190] Especially advantageous polylactic acids for the purposes of the present invention are poly-D-, poly-L- or poly-D, L-lactic acids in particular.

[0191] The expression “polylactic acid” generally refers to homopolymers of lactic acid such as z. y (L-lactic acid), poly(D-lactic acid), poly(DL-lactic acid), mixtures thereof and copolymers which contain lactic acid as the primary component and a small proportion, preferably less than 10% molar, of a co-polymerizable co-monomer.

[0192] Further suitable materials are copolymers or terpolymers based on polylactic acid,

polyglycolic acid, polyalkylene carbonates (such as polyethylene carbonate), polyhydroxyalkanoates (PHA), polyhydroxybutyrates (PHB), polyhydroxyvalerates (PHV) and polyhydroxybutyrate-hydroxyvalerate copolymers (PHBV).

[0193] In a particularly preferred embodiment, the biopolymer is exclusively a thermoplastic polycondensate based on lactic acids.

[0194] The polylactic acids used in accordance with the invention preferably have a number average molecular weight (M_n) which is a minimum of 500 g/mol, preferably a minimum of 1000 g/mol, particularly preferably a minimum of 5000 g/mol, appropriately a minimum of 10000 g/mol, in particular a minimum of 25000 g/mol. On the other hand, the number average is preferably a maximum of 1000000 g/mol, appropriately a maximum of 500000 g/mol, advantageously a maximum of 100000 g/mol, in particular a maximum of 50000 g/mol. A number average molecular weight in the range from a minimum of 10000 g/mol to 500000 g/mol has proved to be particularly advantageous in the context of the present invention.

[0195] The mass average molecular weight (M_w) of preferred lactic acid polymers, in particular of poly-D-, poly-L- or poly-D,L-lactic acids, is preferably in the range 750 g/mol to 5000000 g/mol, preferably in the range 5000 g/mol to 1000000 g/mol, particularly preferably in the range 10000 g/mol to 500000 g/mol, in particular in the range 30000 g/mol to 500000 g/mol, and the polydispersity of these polymers is advantageously in the range 1.5 to 5.

[0196] The inherent viscosity of particularly suitable lactic acid polymers, poly-D-, poly-L- or poly-D,L-lactic acids in particular, measured in chloroform at 25° C., 0.1% polymer concentration, is in the range 0.5 dl/g to 8.0 dl/g, preferably in the range 0.8 dl/g to 7.0 dl/g, in particular in the range 1.5 dl/g to 3.2 dl/g.

[0197] Furthermore, the inherent viscosity of particularly suitable lactic acid polymers, in particular poly-D-, poly-L- or poly-D,L-lactic acids, measured in hexafluoro-2-propanol at 30° C., at 0.1% polymer concentration, is in the range from 1.0 dl/g to 2.6 dl/g, in particular in the range from 1.3 dl/g to 2.3 dl/g.

[0198] Of particular interest are polylactic acids with a glass transition temperature between 50° C. and 65° C.

[0199] Of particular interest are polylactic acids with a melting point between 155° C. and 180° C.

[0200] Of particular interest are commercially available polylactic acids such as NatureWorks PLA 6202D.

[0201] The term “polyhydroxy fatty acid esters” (PHF) as used in the context of the invention should be understood to mean the following polymers: poly(3-hydroxypropionate) (PHP), poly(3-hydroxybutyrate) (PHB, P3HB), poly(3-hydroxyvalerate) (PHV), poly(3-hydroxyhexanoate) (PHHx), poly(3-hydroxyheptanoate) (PHH), poly(3-hydroxyoctanoate) (PHO), poly(3-hydroxynonanoate) (PHN), poly(3-hydroxydecanoate) (PHD), poly(3-hydroxyundecanoate) (PHUD), poly(3-hydroxydodecanoate) (PHDD), poly(3-hydroxytetradecanoate) (PHTD), poly(3-hydroxypentadecanoate) (PHPD), poly(3-hydroxyhexadecanoate) (PHHxD) as well as blends of the aforementioned polymers. In addition to the aforementioned homopolymers, polyhydroxy fatty acid ester copolymers such as poly(3-hydroxypropionate-co-3-hydroxybutyrate) (P3HP-3HB), poly(3-hydroxypropionate-co-4-hydroxybutyrate) (P3HP-4HB), poly(3-hydroxybutyrate-co-4-hydroxybutyrate) (P(3HB-4HB)), poly(3-hydroxybutyrate-co-3-hydroxyvalerate) (PHBV), poly(3-hydroxybutyrate-co-3-hydroxyvalerate-co-3-hydroxyhexanoate) (PHBV-HHx) as well as blends of the aforementioned copolymers, may be used together or with the aforementioned homopolymers.

[0202] The thermoplastic polyhydroxy fatty acid ester polymers used in accordance with the invention are commercially available; examples are Mirel, Biomer P 209, Biopol, Aonilex X, Proganic.

[0203] The thermoplastic polyhydroxy fatty acid ester polymers used in accordance with the invention preferably have a glass transition temperature in the range -2° C. to 62° C.

[0204] The thermoplastic polyhydroxy fatty acid ester polymers used in accordance with the

invention preferably have a melting temperature in the range 100° C. to 177° C.

[0205] The thermoplastic polyhydroxy fatty acid ester polymers used in accordance with the invention preferably have a melt flow index (MFI) of 5-10 g/10 min (190° C., 2.16 kg) determined in accordance with ISO 1133-1:2011.

[0206] The thermoplastic polyhydroxy fatty acid ester polymers used in accordance with the invention preferably have a number average molecular weight (Mn) of at least 200000 Dalton, in particular at least 220000 Dalton, particularly preferably at least 250000 Dalton, and a maximum of up to 3000000 Dalton, in particular up to 2500000 Dalton, particularly preferably up to 2000000 Dalton.

[0207] The thermoplastic polyhydroxy fatty acid ester polymers used in accordance with the invention usually have a mass average molecular weight (Mw) which is about a factor of 2, preferably a factor of 3, times the number average molecular weight (Mn).

[0208] The term “succinate-based aliphatic polymers” should be understood to mean polymers with the following general formula:

##STR00002##

wherein R.sub.1, R.sub.2, R.sub.3, R.sub.4 represent linear or branched aliphatic hydrocarbon residues consisting of 2 to 20 carbon atoms.

[0209] Examples in this regard are polybutylene succinate, polybutylene succinate adipate and polyethylene succinate.

[0210] The thermoplastic succinate-based aliphatic polymers used in accordance with the invention are commercially available; examples are Bionolle 1000, BioPBS.

[0211] The thermoplastic succinate polymers used in accordance with the invention preferably have a glass transition temperature in the range -45° C. to 45° C.

[0212] The thermoplastic succinate polymers used in accordance with the invention preferably have a crystallization temperature in the range 70° C. to 90° C.

[0213] The thermoplastic succinate polymers used in accordance with the invention preferably have a melting temperature in the range 60° C. to 180° C.

[0214] The thermoplastic succinate polymers used in accordance with the invention preferably have a melt flow index (MFI) of 5-10 g/10 min (190° C., 2.16 kg), determined in accordance with ISO 1133-1:2011.

[0215] The thermoplastic succinate polymers used in accordance with the invention preferably have a number average molecular weight (Mn) of at least 20000 Dalton, in particular at least 30000 Dalton, particularly preferably at least 35000 Dalton, and a maximum of up to 140000 Dalton, in particular up to 120000 Dalton, particularly preferably up to 110000 Dalton.

[0216] The thermoplastic succinate polymers used in accordance with the invention usually have a mass average molecular weight (Mw) which is about a factor of 2, preferably a factor of 3, times the number average molecular weight (Mn).

[0217] Polycaprolactone (PCL) is a synthetic biopolymer within the meaning of the present invention.

[0218] Of particular interest are polycaprolactones with a glass transition temperature between -45° C. and 45° C.

[0219] Of particular interest are polycaprolactones with a crystallization temperature between 70° C. and 90° C.

[0220] Of particular interest are polycaprolactones with a melting point between 60° C. and 180° C.

[0221] Of particular interest are polycaprolactones with a melting enthalpy of 70-145 J/g.

[0222] Of particular interest are polycaprolactones with a number average molecular weight (Mn), preferably determined by gel permeation chromatography against polystyrene standards with a narrow distribution or by end group titration, of at least 20000 Dalton to 140000 Dalton.

[0223] Of particular interest are commercially available polycaprolactones such as Resomer C 209.

Thermoplastic Polymer B

[0224] The thermoplastic polymer B is selected from the aforementioned group of thermoplastic polymers and preferred embodiments of thermoplastic polymer B correspond to the preferred embodiments of thermoplastic polymer A, as described above.

[0225] In a preferred embodiment, at least the thermoplastic polymer A and/or the thermoplastic polymer B is/are selected from the group formed by the melt spinnable synthetic biopolymer, wherein polycondensates and polymerisates from bio-based starting materials are particularly preferred. Insofar as both thermoplastic polymers A and B are selected from the group formed by melt spinnable synthetic biopolymers, it is preferable to select biopolymers which differ as regards their chemical nature and/or as regards their melting points. In this embodiment, the multi-component polymer fibres are preferably bi-component fibres in which the component A forms the core and the component B forms the shell. Particularly preferably, the melting point of the thermoplastic polymer in the component A is at least 5° C., preferably at least 10° C., higher than the melting point of the thermoplastic polymer in the component B.

Additives A and B

[0226] The additives A and B increase the biological degradability of the multi-component polymer fibres in accordance with the invention, in particular of the bi-component fibres in accordance with the invention, in that these additives increase the biological degradability of the thermoplastic polymer A and/or of the thermoplastic polymer B.

[0227] The multi-component polymer fibre in accordance with the invention, in particular the preferred bi-component fibre, contains (i) at least one additive A in the component A, or (ii) at least one additive B in the component B, or (iii) at least one additive A in the component A and at least one additive B in the component B. When at least one additive A is present in the component A and at least one additive B is present in the component B, the additive A and the additive B are different, or when at least one additive A is present in the component A and at least one additive B is present in the component B, the additive A and the additive B may also be identical, if the thermoplastic polymer A and thermoplastic polymer B are different. The term “different” in the context of this paragraph means that the substances differ at least as regards their chemical natures or as regards their physical natures or as regards their concentrations.

[0228] In particular, the additives A and B are: [0229] basic alkali and/or alkaline earth compounds (pH>7 dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly preferably CaCO₃, and alkaline additives, particularly preferably CaO [0230] aliphatic polyesters, preferably aliphatic polyesters having no side chain carbon atoms, preferably polycaprolactone [0231] fatty acid ester, preferably C1-C40-alkyl stearate, more preferred C2-C20-alkyl stearate, most preferred ethyl stearate [0232] sugars, in particular monosaccharides, disaccharides and oligosaccharides [0233] catalysts for transesterifications, in particular under basic conditions [0234] metal compounds, in particular transition metal compounds, preferably at least two transition metal compounds, as well as their salts [0235] unsaturated carboxylic acids or their anhydrides/esters/amides [0236] synthetic rubber, natural rubber [0237] carbohydrates, in particular starch and/or cellulose as well as mixtures of the aforementioned substances.

[0238] Multi-component polymer fibres, in particular bi-component polymer fibres, in which the thermoplastic polymer A and/or the thermoplastic polymer B comprise(s) at least one polyester and the additive A and/or additive B is/are selected from the group (i) basic alkali and/or alkaline earth compounds (pH>7 dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly preferably CaCO₃, and alkaline additives, particularly preferably CaO, (ii) aliphatic polyester, (iii) fatty acid ester, preferably C1-C40-alkyl stearate, more preferred C2-C20-alkyl stearate, most preferred ethyl stearate, (iv) sugars, in particular monosaccharides, disaccharides and oligosaccharides, (v) catalysts for transesterifications, in particular under basic conditions, (vi) carbohydrates, in particular starch and/or cellulose, as well as mixtures thereof, are preferred.

[0239] Particularly preferred bi-component polymer fibres are those in which the thermoplastic polymer A and/or the thermoplastic polymer B comprises at least one polyester and the additive A and/or additive B is selected from the group (i) basic alkali and/or alkaline earth compounds (pH>7 dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly preferably CaCO_3 , and alkaline additives, particularly preferably CaO, (ii) aliphatic polyesters, (iii) fatty acid ester, preferably C1-C40-alkyl stearate, more preferred C2-C20-alkyl stearate, most preferred ethyl stearate, (iv) sugars, in particular mono-saccharides, di-saccharides and oligo-saccharides, (v) catalysts for transesterifications, in particular under basic conditions, (vi) carbohydrates, in particular starch and/or cellulose, as well as mixtures thereof. The aforementioned aliphatic polyesters are distinguished from the polyesters of the thermoplastic polymer A and polymer B in respect of their chemical nature, i.e. the polyester of the thermoplastic polymer A and polymer B is an araliphatic polyester or copolyester, which has been produced from polyols and aliphatic and/or aromatic dicarboxylic acids or their derivatives (anhydrides, esters) by means of polycondensation.

[0240] Particularly preferred additives A and/or additives B contain at least two substances, wherein preferred combinations are: [0241] A) basic alkali and/or alkaline earth compounds (pH>7 dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly preferably CaCO_3 , and alkaline additives, particularly preferably CaO in combination with catalysts for transesterifications, in particular under basic conditions; [0242] B) sugars, in particular mono-saccharides, di-saccharides and oligo-saccharides, in combination with carbohydrates, in particular starch and/or cellulose, as well as mixtures thereof; [0243] C) aliphatic polyesters, optionally in combination with sugars, in particular mono-saccharides, di-saccharides and oligo-saccharides, or carbohydrates, in particular starch and/or cellulose, as well as mixtures thereof; [0244] D) fatty acid ester, preferably C1-C40-alkyl stearate, more preferred C2-C20-alkyl stearate, most preferred ethyl stearate.

[0245] Most preferred additives A for partially aromatic “araliphatic” polyester or copolyester as thermoplastic polymer A are containing at least [0246] basic alkali and/or alkaline earth compounds (pH>7 dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly preferably CaCO_3 , and alkaline additives, particularly preferably CaO in combination with catalysts for transesterifications, in particular under basic conditions; and [0247] aliphatic polyesters, especially aliphatic polyester having no side chain carbon atoms, optionally in combination with (i) sugars, in particular mono-saccharides, di-saccharides and oligo-saccharides, (ii) carbohydrates, in particular starch and/or (iii) cellulose, (iv) fatty acid ester, preferably C1-C40-alkyl stearate, more preferred C2-C20-alkyl stearate, most preferred ethyl stearate, as well as mixtures thereof.

[0248] Out of the aforementioned particularly preferred bi-component polymer fibres are those preferred in which the thermoplastic polymer A is polyester and the thermoplastic polymer B is a polyester being different from the polyester in polymer A, and preferably is a co-polyester, and—each—the additive A and the additive B is independently selected from the combination of [0249] basic alkali and/or alkaline earth compounds (pH>7 dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly preferably CaCO_3 , and alkaline additives, particularly preferably CaO in combination with catalysts for transesterifications, in particular under basic conditions;

and [0250] aliphatic polyesters, especially aliphatic polyester having no side chain carbon atoms, optionally in combination with (i) sugars, in particular mono-saccharides, di-saccharides and oligo-saccharides, (ii) carbohydrates, in particular starch and/or (iii) cellulose, (iv) fatty acid ester, preferably C1-C40-alkyl stearate, more preferred C2-C20-alkyl stearate, most preferred ethyl stearate, as well as mixtures thereof.

[0251] In a particular preferred embodiment, the aforementioned fatty acid esters are present and not optionally.

[0252] Multi-component polymer fibres, in particular bi-component polymer fibres, in which the thermoplastic polymer A and/or the thermoplastic polymer B comprise(s) at least one polyolefin and the additive A and/or additive B is/are selected from the group (i) sugars, in particular monosaccharides, disaccharides and oligosaccharides, (ii) metal compounds, in particular transition metal compounds, as well as their salts, (iii) unsaturated carboxylic acids or their anhydrides/esters/amides, (iv) synthetic rubber and/or natural rubber, (v) carbohydrates, in particular starch and/or cellulose, as well as mixtures thereof, are preferred. In particular preferred for polyolefin is additive A and/or additive B comprising (a) transition metal compounds and (b) unsaturated carboxylic acids or their anhydrides, which are in particular preferred combined with (c) synthetic rubber and/or natural rubber and (d) starch.

[0253] Particularly preferred bi-component polymer fibres are those in which the thermoplastic polymer B is a polyolefin, in particular a polypropylene polymer, which includes as additive B at least (i) metal compounds, in particular transition metal compounds, as well as their salts, preferably at least two chemically different transition metal compounds and (ii) unsaturated carboxylic acids or their anhydrides/esters/amides, preferably in combination with synthetic rubber and/or natural rubber, and—optionally—further comprising (iii) sugars, in particular monosaccharides, disaccharides and oligosaccharides, (iv) carbohydrates, in particular starch and/or (v) cellulose, as well as mixtures thereof. Further, phenolic antioxidant stabilizer and CaO can be present.

[0254] Multi-component polymer fibres, in particular bi-component polymer fibres, in which the thermoplastic polymer A and/or the thermoplastic polymer B comprise(s) at least one polyamide and the additive A and/or additive B is/are selected from the group (i) basic alkali and/or alkaline earth compounds (pH>7 dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly preferably CaCO_3 , and alkaline additives, particularly preferably CaO, (ii) aliphatic polyester, (iii) fatty acid ester, preferably C1-C40-alkyl stearate, more preferred C2-C20-alkyl stearate, most preferred ethyl stearate, (iv) sugars, in particular monosaccharides, disaccharides and oligosaccharides, (v) catalysts for transesterifications, in particular under basic conditions, (vi) metal compounds, in particular transition metal compounds, as well as their salts, (vii) unsaturated carboxylic acids or their anhydrides/esters/amides, (viii) synthetic rubber and/or natural rubber, (ix) carbohydrates, in particular starch and/or cellulose, as well as mixtures thereof, are preferred.

[0255] The additive A is in a proportion with respect to the component A which is preferably between 0.005% by weight and 20% by weight, particularly preferably between 0.01% by weight and 5% by weight, with respect to the total weight of the component A.

[0256] The additive B is in a proportion with respect to the component B which is preferably between 0.005% by weight and 20% by weight, particularly preferably between 0.01% by weight and 5% by weight, with respect to the total weight of the component B.

[0257] In order to obtain a low proportion by weight as well as a distribution of the additive in the component, which is as uniform as possible, the additives are preferably added to the polymer material in the extruder in the form of what is known as a masterbatch.

[0258] The term “masterbatch” should be understood to mean a granulate which is added to the polymer melt during the spinning process. In this regard, the granulate has a polymeric support material as well as at least one additive.

[0259] In order to enable small quantities of additive to be added to the polymer, preferably, the concentration of the additive/additives in the masterbatch is preferably tailored. Preferably, the dosage of masterbatch in the spinning process is between 0.1% by weight and 30% by weight, particularly preferably between 0.5% by weight and 15% by weight.

Thermobonding

[0260] For thermobonding, suitable thermoplastic polymers, copolymers and blends in particular, in particular thermoplastic biopolymers are those which have a high degree of enthalpies of fusion

and crystallization. Usually, the polymers B are selected in a manner such that they have a degree of crystallinity or a latent heat of fusion (ΔH_f) of more than approximately 25 joules per gram ("J/g"), particularly preferably more than 35 J/g, in particular more than 50 J/g. The determination of the latent heat of melting (ΔH_f), the latent heat of crystallization (ΔH_c) and the crystallization temperature is carried out by means of Differential Scanning Calorimetry ("DSC") in particular in accordance with ASTM D-3418 (ASTM D3418-15, Standard Test Method for Transition Temperatures and Enthalpies of Fusion and Crystallization of Polymers by Differential Scanning Calorimetry, ASTM International, West Conshohocken, PA, 2015, www.astm.org).

Further Additives to Thermoplastics a and B

[0261] The thermoplastic polymers, copolymers and blends described above, in particular the biopolymers described above, have the usual additives such as antioxidants, inter alia.

[0262] Further usual additives are pigments, stabilizers, surfactants, waxes, flow promoters, solid solvents, plasticizers and other materials, for example nucleating agents, which are added in order to improve the processability of the thermoplastic composition.

[0263] The multi-component fibres in accordance with the invention, in particular the bi-component fibres in accordance with the invention, are constituted by at least 90% by weight of thermoplastic polymers, copolymers, blends described above, in particular of thermoplastic biopolymers, and typically have less than approximately 10% by weight, preferably less than approximately 8% by weight, particularly preferably less than approximately 5% by weight of additives, in particular in the shell.

[0264] The multi-component fibres in accordance with the invention, in particular the bi-component fibres in accordance with the invention, may be continuous fibres, for example what are known as staple fibres, or continuous fibres (filaments).

Production of Multi-Component Fibres

[0265] After spinning into tow, the multi-component fibres, in particular the bi-component fibre in accordance with the invention, are combined together and post-treated in a rolling mill using methods which are known in principle, in particular drawn and optionally also crimped or texturized.

[0266] When processed after the (filament) spinning process, the multi-component polymer fibres in accordance with the invention are cooled immediately after exiting the spinneret and drawn and deposited on a collecting belt or preferably wound onto bobbins. Further steps in particular include drawing, texturizing and heat bonding of the filaments.

[0267] The production of the multi-component fibres in accordance with the invention, in particular of the bi-component fibres in accordance with the invention, is carried out using methods and equipment which is known to the person skilled in the art, and these have been described in the literature, for example in Fourné (Synthetische Fasern [Synthetic Fibres]; 1995, Chapter 4 and Chapter 5.2).

[0268] A number of production methods are available for the production of nonwovens. In the production of spunbonds, the intermediate step of staple fibre production is not carried out. In particular, the multi-component fibres are swirled directly after exiting the spinnerets, preferably by means of a stream of air, so that they are deposited as a nonwoven. The production of spunbonds is known to the person skilled in the art and has been described in the literature, for example in Fourné (Synthetische Fasern [Synthetic Fibres]; 1995, Chapter 5.5).

[0269] In order to improve the dispersibility or for the purposes of further processing in the secondary spinning unit, in particular into yarns, the fibre is preferably in the form of a staple fibre. The length of said staple fibres is not limited in principle, but in general is 2 to 200 mm, preferably 3 to 120 mm, particularly preferably 4 to 60 mm.

[0270] The individual linear density of the multi-component fibres in accordance with the invention, in particular of the bi-component fibres in accordance with the invention, preferably staple fibres, is between 0.5 and 30 dtex, preferably 0.7 to 13 dtex. For some applications, linear

densities between 0.5 and 3 dtex and fibre lengths of <10 mm, in particular <8 mm, particularly preferably <6 mm, particularly preferably <5 mm, are particularly suitable.

[0271] The multi-component fibres in accordance with the invention, in particular the bi-component fibres in accordance with the invention, preferably have a low hot air heat shrinkage in the range 0% to 10%, preferably >0% to 8%, respectively measured at 110° C.

[0272] The production of the polymer fibres in accordance with the invention is in principle carried out using the usual processes. Firstly, the polymer, if necessary, is dried and supplied to an extruder. Next, the molten material is spun using the regular equipment with appropriate spinnerets. The mass throughput and the draw-off speed of the capillaries from the spinneret outlet plates are set so that a fibre with the desired linear density is produced.

[0273] The fibres formed may have different shapes, for example round, oval, star-shaped, dog-bone shaped, barbell-shaped, kidney-shaped, triangular or polygonal, cloverleaf-shaped, horseshoe-shaped, lens-shaped, rod-shaped, gearwheel-shaped, cloud-shaped, x-shaped, y-shaped, o-shaped, u-shaped; this list is not limiting and other suitable cross sections are also possible.

[0274] The fibre filaments produced in accordance with the invention are collected into yarns and then in turn into tows. The tows are initially deposited into cans for further processing. The tows which are temporarily stored in the cans are picked up and a large cable tow is produced.

[0275] The present invention also concerns the post-treatment of the cable tow produced by means of the known process; usually, it is 10-600 ktex using conventional rolling mills, and special drawing. An infeed speed for the cable tow into the drawing or drawing equipment is preferably 10 to 110 m/min (infeed speed). In this regard, other preparations may also be applied which aid drawing but which do not have a deleterious effect on the subsequent properties.

[0276] Drawing may be carried out in a single step or optionally using a two-stage drawing process (see in this regard U.S. Pat. No. 3,816,486, for example). Prior to and during drawing, one or more finishing agents may be applied using conventional methods.

[0277] The drawing in accordance with the invention is carried out with a draw ratio, in particular when biopolymers are used, of between 1.2 and 6.0, preferably between 2.0 and 4.0, wherein the temperature when drawing the tow is preferably between 30° C. and 100° C. Drawing is thus carried out in the glass transition temperature range for the tow to be drawn. Drawing in accordance with the invention is carried out in the presence of steam, i.e. in what are known as steam boxes, so that the fibres are drawn in the steam boxes. The steam boxes are normally operated under a pressure of 3 bar.

[0278] By drawing in the presence of steam in the aforementioned temperature range, thermal shrinkage of the fibres can be reduced and controlled in a specific manner.

[0279] The tow is preferably 24-360 ktex prior to drawing.

[0280] Drawing is preferably in one stage or in multiple stages, wherein the godets of the drawing unit may be at different temperatures and also the draw ratios between the drawing unit may be different. Preferably, a steam box is positioned between at least two of the drawing units, i.e. the drawing point for the fibres is in the steam box or close to the steam box. All of the godets (usually 7 per drawing unit) are at a temperature of 30-250° C. All of the drawing is preferably carried out at least partially or entirely in the steam box. Preferably, the steam box is operated at a pressure of 3 bar of steam.

[0281] Drawing may also be carried out cold, wherein "cold" means room temperature (approximately 20-35° C.).

[0282] Carrying out of the respective drawing as well as the choice of all of the parameters for the rolling mill is carried out as a function of the polymer and/or the end use of the fibres.

[0283] For the optional crimping/texturizing of the drawn fibres, conventional methods of mechanical crimping with crimping machines which are known per se may be used. Preferably, a mechanical device for fibre crimping with steam support is used, such as a stuffer box. However, crimped fibres may be obtained using other processes, including three-dimensionally crimped

fibres, for example. In order to carry out the crimping, the tow is initially and usually brought to a constant temperature in the range 50° C. to 100° C., preferably 70° C. to 85° C., particularly preferably to approximately 78° C. and treated at a pressure for the tow infeed rolls of 1.0 to 6.0 bar, particularly preferably at approximately 2.0 bar, a pressure in the crimping box of 0.5 to 6.0 bar, particularly preferably 1.5-3.0 bar, with steam at a rate of between 1.0 and 2.0 kg/min, particularly preferably 1.5 kg/min.

[0284] If the smooth or optionally crimped fibres are relaxed and/or fixed in an oven or stream of hot air, this is also carried out at maximum temperatures of 130° C.

[0285] In order to produce staple fibres, the smooth or optionally crimped fibres are picked up, followed by cutting and depositing into compressed bales as flock. The staple fibres of the present invention are preferably cut on a mechanical cutting device which is downstream of relaxation. In order to produce different types of tow, cutting may be dispensed with. These types of tow are deposited and compressed in the uncut form in bales.

[0286] When the fibres in accordance with the invention are in a crimped embodiment, then the degree of crimping is preferably at least 2 crimps (arched crimps) per cm, preferably at least 3 crimps per cm, preferably 3 crimps per cm to 9.8 crimps per cm and particularly preferably 3.9 crimps per cm to 8.9 crimps per cm. In applications for the production of textile fabrics, values for the degree of crimping of approximately 5 to 5.5 crimps per cm are particularly preferred. For the production of textile fabrics using wet laid processes, the degree of crimping has to be set individually.

[0287] A typical set-up for producing bi-component fibre of the core/shell (=core/sheath) type having polyethylene terephthalate (PET) as thermoplastic polymer A and additive A in the core and polypropylene (PP) as thermoplastic polymer B and additive B in the shell (sheath) includes the following: [0288] The PET raw material is dried, typically up to 4-6 h @ temp up to 180° C.; Typically, polypropylene (PP) does not require drying; [0289] The melt extrusion is typically done extruders having one or more screws; [0290] The bicomponent spinneret configuration is concentric or eccentric with PP as shell (sheath) material and PET as core component; [0291] The extruder melt temperatures for core is typically in the range 250-300° C. for PET and for sheath material typically in the range 220-270° C. for PP; [0292] Additives added at extruder feed-throat for both shell (sheath) and core at a level between 1-3 wt.-%, typically in the form of a masterbatch; [0293] Fibre Quench is typically crossflow and the air temperature is typically in the range 18-24° C.; [0294] typical fibre drawdown speeds are in the range 800-1300 m/min; [0295] fibre drawing can be single or duo-stage drawing with draw ratio up to 4 and heat setting at 110-130° C.

[0296] A typical set-up for producing bi-component fibre of the core/shell (=core/sheath) type having polyethylene terephthalate polymer (PET) as thermoplastic polymer A and additive A in the core and polyethylene terephthalate copolymer (coPET) as thermoplastic polymer B and additive B in the shell (sheath) includes the following: [0297] =The PET raw materials are dried, typically up to 4-6 h @ temp up to 180° C.; [0298] The melt extrusion is typically done extruders having one or more screws, one extruder for shell (sheath) material (coPET) and one for core material (PET); [0299] The bicomponent spinneret configuration is concentric or eccentric with coPET as shell (sheath) material and PET as core component; [0300] The extruder melt temperatures are typically in the range 250-300° C.; [0301] Additives added at extruder feed-throat for both shell (sheath) and core at a level between 1-3 wt.-%, typically in the form of a masterbatch; [0302] Fibre quench is typically crossflow or in-flow or radial out-flow and the air temperature is typically in the range 18-50° C.; [0303] typical fibre drawdown speeds are in the range 400-1800 m/min, preferably 1400 m/min; [0304] fibre drawing can be single or duo-stage drawing with draw ratio up to 4.5, specifically 2.5-3.5, finish bath temperature up to 80° C., godet temperatures up to 70° C., specifically 30° C. before steam-bath if present, and temperature after stretch point up to 80° C., heat setting, typically in a hot air oven, at temperatures up to 190° C.

[0305] Textile fabrics can be produced from the fibres in accordance with the invention; these also constitute the subject matter of the invention.

Textile Fabric

[0306] The term “textile fabric” as used in the context of this description should be construed in its broadest sense. Thus, they may be any structure containing the fibres in accordance with the invention which have been produced using a technique for producing a fabric. Examples of such textile fabrics are nonwovens, in particular wet laid nonwovens or dry laid nonwovens, preferably based on staple fibres which are produced by means of thermobonding. Other examples of nonwovens are carded or airlaid nonwovens, preferably based on staple fibres or nonwovens, produced using a melt blowing and/or spunbond filament process. Particularly in the case in which the fibres or nonwovens have a low linear density, meltblown processes (for example as described in “Complete Textile Glossary”, Celanese Acetate LLC, from 2000 or in “Chemiefaser-Lexikon, Robert Bauer, 10th edition, 1993) and electrospinning processes are the most suitable.

[0307] In order to produce a nonwoven using the spunbond filament process, the freshly spun fibres, preferably freshly spun bi-component fibres, are collected on a collecting conveyor to lay them up to a specified thickness so that the spunbond nonwoven can be obtained. The spunbond nonwoven can be consolidated further, for example using the hot embossing process with the use of an embossing roller or using known needling/water jet processes, to further entangle the nonwoven. When bi-component fibres are used, wherein the bi-component fibres have a higher and a lower melting point component, the nonwoven is consolidated by means of thermobonding using the lower melting point component.

[0308] For the aforementioned thermobonding, the textile fabric which contains the bi-component/multi-component fibres, is fed into an oven, for example a ventilated dryer which contains one or more heating zones which are used to heat the air to a temperature which is higher than the melting temperature of the lower melting point component (for example the shell) of the multi-component fibres, but lower than the melting temperature of the higher melting point component (for example the core). This heated air flows through the textile fabric, typically a nonwoven, whereupon the lower melting point component melts and forms bonds between the fibres in order to stabilize the fabric thermally.

[0309] Typically, the air flowing through the thermobonding oven is at a temperature in the range from 100° C. to approximately 180° C. The residence time in the oven is approximately 180 seconds or less. It should be understood, however, that the parameters of the thermobonding oven are a function of the type of polymers used and the thickness of the material.

[0310] Ultrasound consolidation techniques may also be used, which employ a stationary or rotating horn and a rotating patterned embossing roller. Examples of such techniques are described in U.S. Pat. Nos. 3,939,033; 3,844,869; 4,259,399; 5,096,532; 5,110,403 and 5,817,199, which are incorporated herein by reference in their entirety for all purposes. As an alternative, the nonwoven may be thermally spot welded in order to provide a fabric with a great many small, discrete binding points. This process in general involves guiding the fabric between two heated rollers such as, for example, a roller with an engraved pattern and a second binding roller. The engraved roller is patterned in a manner such that the web is not bonded over its entire surface, and the second roller may be smooth or patterned.

[0311] For functional and/or aesthetic reasons, a variety of patterns have been developed for engraved rollers. Examples of binding patterns include but are not limited to those described in U.S. Pat. Nos. 3,855,046; 5,620,779; 5,962,112; 6,093,665; U.S. design Pat. Nos. 428,267 and 390,708, which are incorporated herein by reference in their entirety for all purposes.

[0312] The basis weight of the textile fabric, in particular the basis weight of the nonwoven, is between 10 and 500 g/m.², preferably 25 to 450 g/m.², in particular 30 to 300 g/m.².

[0313] The textile fabric which is produced from the multi-component fibres in accordance with the invention, in particular from the bi-component fibres in accordance with the invention, in

particular nonwovens, can be produced in a known manner using a calendar roller or can be thermally consolidated in an oven.

[0314] The textile fabric which is produced from the multi-component fibres in accordance with the invention, for example nonwovens, are usually produced by means of thermobonding because of the different melting points of the components. This bonds the fibres together at the contact or crossover points. Insofar as the component B produced from thermoplastic polymer B with additive B has a higher biological degradability than the component A produced from thermoplastic polymer A with additive A, the contact or crossover points of the fibres with each other are degraded first and the textile fabric, for example a nonwoven, disintegrates faster, whereupon the overall degradability is increased.

[0315] Here, the textile fabrics, in particular the nonwovens, can—in addition to the multi-component fibres—comprise still other fibres, depending on the intended purpose. In this regard, the “filler fibres” described in WO 2007/107906 should in particular be highlighted. The “filler fibres” described in WO 2007/107906 also form part of the subject matter of the invention and are incorporated into the present invention.

[0316] The textile fabrics include the aforementioned biologically degradable polymer material fibres which may be mixed with other fibrous materials, chemical fibres, preferably natural fibres such as cotton or cellulose fibres, fibres of animal origin such as wool or other biologically degradable fibres. When mixing such different fibres, textile fabrics with a fibre gradient may be produced. Examples of cellulose fibres include softwood kraft pulp fibres. Softwood kraft pulp fibres are obtained from conifers and include cellulose fibres such as, but not restricted to, northern, western and southern softwood species such as redwood, red cedar, hemlock spruce, Douglas fir, true spruces, pine trees (for example southern pine), spruce (for example black spruce), combinations thereof etc. In the present invention, northern softwood kraft pulp fibres may be used. Another suitable cellulose material for use in the present invention is a bleached sulphate wood cellulose material which primarily contains softwood fibres. Fibres with smaller average lengths may also be used in the present invention. An example of suitable cellulose material fibre with a low average length are hardwood kraft pulp fibres. Hardwood kraft pulp fibres are derived from deciduous trees and include cellulose material fibres such as, but not restricted to *eucalyptus*, maple, beech, aspen etc. *Eucalyptus* kraft pulp fibres may be particularly favoured in order to increase softness, increase sheen, increase opacity and change the pore structure of the sheet in order to increase its absorbency. Typically, cellulose material fibres make up approximately 30% by weight to approximately 95% by weight, in some embodiments approximately 40% by weight to approximately 90% by weight and in some embodiments approximately 50% by weight to approximately 85% by weight of the nonwoven.

[0317] In addition, superabsorbent materials may also be contained in the nonwoven.

Superabsorbent materials are materials which swell in water which can absorb 20 times their weight and in some case at least 30 times their weight in an aqueous solution containing 0.9% by weight of sodium chloride. The superabsorbent materials may be natural, synthetic and modified natural polymers and materials. Examples of synthetic superabsorbent polymers include alkali metal and ammonium salts of poly(acrylic acid) and poly(methacrylic acid), poly(acrylamides), polyvinylethers), maleic acid anhydride copolymers with vinyl ethers and alpha olefins, polyvinylpyrrolidone), poly(vinylmorpholinone), polyvinylalcohol) and mixtures and copolymers thereof. Other superabsorbent materials include natural and modified natural polymers such as hydrolysed acrylonitrile-grafted starches, acrylic acid-grafted starches, methylcelluloses, chitosan, carboxymethylcelluloses, hydroxypropylcelluloses and natural gums such as alginates, xanthan gums, carob bean gum etc. Mixtures of natural and completely or partially synthetic superabsorbent polymers may also be useful in the present invention. When the superabsorbent material is used, it may make up approximately 30% by weight to approximately 95% by weight, in some embodiments approximately 40% by weight to approximately 90% by weight and in some

embodiments approximately 50% by weight to approximately 85% by weight of the nonwoven.

[0318] The present textile fabrics, in particular the aforementioned nonwoven, may be used in an absorbent article such as, for example, absorbent articles for body care such as, for example, nappies, training pants, absorbent underwear, incontinence articles, sanitary wear for women, but not restricted thereto (for example sanitary towels), swimwear, baby wipes etc; medical absorbent articles such as clothing, window materials, underlays, bed protectors, bandages, absorbent cloths and medical wipes; wipes for the food industry; items of clothing, etc. Materials and processes which are suitable for the production of absorbent articles of this type are known to the person skilled in the art. Typically, absorbent articles comprise a substantially liquid-impermeable layer (for example the outer shell), a liquid-permeable layer (for example the layer facing the body), barrier layer etc) and an absorbent core. The nonwoven in the present invention may be used as one or more of the liquid-impermeable, liquid-permeable and/or absorbent layers.

[0319] The present textile fabrics, in particular the nonwoven described above, are not limited to the aforementioned applications and may be used in any application such as, for example, in hygiene, medicine, personal protection, in the household (fibre fill etc), clothing, mobility/transport (car, train, aircraft, shipping), engineering (insulation), agriculture, packaging, filtration and any disposable applications.

Test Methods:

[0320] Unless stated otherwise in the present description, the following measurement or test methods were employed:

Linear Density:

[0321] The determination of the linear density was carried out in accordance with DIN EN ISO1973.

Biological Degradability

[0322] The determinations, testing and specifications were in accordance with at least one method selected from the group formed by (i) ASTM D5338-15 (2021) (Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials Under Controlled Composting Conditions, Incorporating Thermophilic Temperatures (DOI: 10.1520/D5338-15R21) ASTM International, West Conshohocken, PA, 2015, www.astm.org), (ii) ASTM D6400-12 (Standard Specification for Labeling of Plastics Designed to be Aerobically Composted in Municipal or Industrial Facilities) (DOI: 10.1520/D6400-12), (iii) ASTM D5511 (ASTM D5511-11 Standard Test Method for Determining Anaerobic Biodegradation of Plastic Materials Under High-Solids Anaerobic digestion Conditions (DOI: 10.1520/D5511-11) and ASTM D5511-18 Standard Test Method for Determining Anaerobic Biodegradation of Plastic Materials Under High-Solids Anaerobic-digestion Conditions; (DOI: 10.1520/D5511-18)), (iv) ASTM D6691 (ASTM D6691-09 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in the Marine Environment by a Defined Microbial Consortium or Natural Sea Water Inoculum) (DOI: 10.1520/D6691-09) and ASTM D6691-17, Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in the Marine Environment by a Defined Microbial Consortium or Natural Sea Water Inoculum (DOI: 10.1520/D6691-17)), (v) ASTM D5210-92 (Anaerobic Degradation in the Presence of Sewage Sludge) (DOI: 10.1520/D5210-92), (vi) PAS 9017:2020 (Plastics—Biodegradation of polyolefins in an open-air terrestrial environment—Specification), ISBN 978 0 539 17478 6; 2021 Oct. 31, (vii) ASTM D5988 (ASTM D5988-12 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in Soil) (DOI: 10.1520/D5988-12), ASTM D5988-18 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in Soil (DOI: 10.1520/D5988-18), ASTM D5988-03 Standard Test Method for Determining Aerobic Biodegradation in Soil of Plastic Materials or Residual Plastic Materials After Composting (DOI: 10.1520/D5988-03)), (viii) EN 13432:2000-12 Packaging—Requirements for packaging recoverable through composting and biodegradation—Test scheme and evaluation criteria for the final acceptance of packaging; German version EN

13432:2000 (DOI: 10.31030/9010637), (ix) ISO 14855-1:2013-04 (DOI: 10.31030/1939267) and ISO 14855-2:2018-07 (ICS 83.080.01) Determination of the ultimate aerobic biodegradability of plastic materials under controlled composting conditions (Method by analysis of evolved carbon dioxide), (x) EN 14995:2007-03—Plastics—Evaluation of compostability (DOI: 10.31030/9730527) or (xi) ISO 17088:2021-04 (Specifications for compostable plastics) (ICS 83.080.01).

Number and Mass Average Molecular Weight (M_n/M_w)

[0323] Determination using gel permeation chromatography against suitable polymer standards with a narrow distribution, in particular DIN 55672 (gel permeation chromatography (GPC)).

Inherent Viscosity

[0324] Determination measured in chloroform at 25° C., 0.1% polymer concentration via GPC.

Glass Transition Temperature and Melting Temperature

[0325] In particular, determination of the glass transition temperature in accordance with DIN EN ISO 11357-2:2020-08 (Plastics—Differential Scanning Calorimetry (DSC)—Part 2: Determination of glass transition temperature and the step height related to glass transition).

[0326] In particular, determination of the melting temperature in accordance with DIN EN ISO 11357-3:2018-07 (Plastics—Differential Scanning Calorimetry (DSC)—Part 3: Determination of the temperatures and enthalpies of melting and crystallization).

[0327] Determination by means of Differential Scanning Calorimetry (DSC) using the following protocol:

[0328] DSC measurement carried out under nitrogen, calibration against indium. Nitrogen flow 50 mL/min; weight of fibres in the range 2-3 mg.

[0329] Temperature range from -50° C. to 210° C. @ 10 K/min then isothermal for 5 min and finally back up to -50° C. @ 10 K/min.

[0330] In general, the final temperature was always approximately 50° C. above the highest expected melting point.

[0331] DSC measurement carried out using a TA/Waters Model Q100.

Melt Viscosity

[0332] The melt viscosity was determined using a Göttfert Rheo Tester 1000 at a temperature suitable for the polymer, between approximately 190° C. and 280° C. In particular, use of ASTM D2196-20 (Standard Test Methods for Rheological Properties of Non-Newtonian Materials by Rotational Viscometer).

Apparent Viscosity

[0333] The determination was carried out as described in WO 2007/070064.

Melt Flow Index

[0334] Determination in accordance with the ASTM test method D1238-13 (ASTM D1238-13, Standard Test Method for Melt Flow Rates of Thermoplastics by Extrusion Plastometer, ASTM International, West Conshohocken, PA, 2013, www.astm.org) or in accordance with DIN EN ISO 1133-1:2012-03 (Plastics—Determination of the melt mass-flow rate (MFR) and the melt volume flow-rate (MVR) of thermoplastics—Part 1: Standard test method) and DIN EN ISO 1133-2:2012-03 (Plastics—Determination of the melt mass-flow rate (MFR) and the melt volume flow-rate (MVR) of thermoplastics—Part 2: Procedure for materials which are sensitive to time-temperature history and/or to moisture). The melt flow index is the weight of a polymer (in grams) that can be pressed through an extrusion rheometer opening (for example 0.0825 inch diameter) when a force of 2160 grams, for example, is applied over a period of 10 minutes, for example, at 190° C., for example.

Latent Heat of Melting

[0335] Determinations of latent heat of melting (ΔH_f), the latent heat of crystallization (ΔH_c) and the crystallization temperature were carried out using Differential Scanning calorimetry (“DSC”) in accordance with ASTM D-3418 (ASTM D3418-15, Standard Test Method for Transition

Temperatures and Enthalpies of Fusion and Crystallization of Polymers by Differential Scanning Calorimetry, ASTM International, West Conshohocken, PA, 2015, www.astm.org) or in accordance with DIN EN ISO 11357 (Plastics-Differential Scanning Calorimetry (DSC)).

Heat Shrinkage

[0336] 12 fibres (test specimens) were prepared from the sample cable tow. They were clamped at one end in a terminal block with the aid of tweezers, and a decrimping weight was fastened to the other end. The measurement was carried out with the aid of a bi-component fibre of the type core/shell with a linear density of 2.2 dtex; the decrimping weight was 190 mg.

[0337] The terminal block with the test specimens was fastened into a support stand so that the test specimens were freely suspended in the support stand under pre-tension. The selected starting length (in normal cases 150 mm) was marked here on each fibre. This was carried out with the aid of marking lines in the support stand and marking points applied to the test specimens. After marking, the filled terminal block was picked up and replaced on the plate. Here, the decrimping weights were removed and the free fibre ends were clamped into a second terminal block. The test specimens spanning the two terminal blocks were suspended in a wire frame, not under tension. This wire frame was introduced into the centre of the shrinkage oven preheated to the correct treatment temperature (usual temperatures are 200° C., 110° C., 80° C.). After the treatment time of 5 min, the wire frame was removed from the oven. After a cooling period for the terminal blocks of at least 30 min, the terminal blocks with the test specimens were removed from the frame and the fibres replaced on the plate. The back measurement could then be carried out. To this end, the test specimens were once again loaded with the decrimping weights and suspended in the support stand. For the back measurement, the adjustable marking line of the support stand was positioned such that the respective upper edge of the marking point could cover the marking line. Then, for each individual fibre, the length between the marks could be read off from the counters on the support stand to an accuracy of 1/10 mm.

[00001] Calculation of change of length:
$$\text{Change of length}[\%] = \frac{\text{initial length}[\text{mm}] - \text{measured length}[\text{mm}]}{\text{initial length}[\text{mm}]} \times 100\%$$

[0338] The average value for all 12 test specimens was used.

[0339] The invention will now be illustrated by the following examples, which does not in any way limit its scope.

Description

EXAMPLE 1

[0340] A polyethylene terephthalate (PET) fibre is spun from a polyethylene terephthalate (PET) resin having the following properties:

[0341] The melt extrusion is done by an extruder having one or more screws at a temperature of 280-290° C. for PET

[0342] Additive A is added at the extruder feed-throat at a level of 2 wt.-% masterbatch dosage. This masterbatch consists of a PET polyester as carrier and the additive, which comprises an aliphatic polyester and CaCO₃.

[0343] The fibre quench occurs by crossflow and air temperature of 40° C.; fibre drawdown speed is 1400 m/min. Spun fiber fineness is 5.4 dtex.

[0344] The fibre drawing is done by single or duo-stage drawing with draw ratio up to 4 and the final dtex is 2.5 dtex. Heat setting is done at 110-130° C.

[0345] The fibre produced is cut into staple fibre having a length of 38 mm.

[0346] The fibre produced is tested in accordance with ASTM D5511 and results are obtained after 208 days:

TABLE-US-00001 3066 - PET Staple Fibre with biodegradable Inculum Negative Positive additive
Cumulative Gas 1366.6 1229.3 9867.0 6356.4 Volume (mL) Percent CH₄ (%) 40.7 31.7 37.4

45.5 Volume CH.sub.4 (mL) 556.0 390.1 3693.0 2890.6 Mass CH.sub.4 (g) 0.40 0.28 2.64 2.06
 Percent CO.sub.2 (%) 41.0 40.5 43.6 38.0 Volume CO.sub.2 (mL) 560.7 498.1 4306.2 2414.8 Mass
 CO.sub.2 (g) 1.10 0.98 8.46 4.74 Sample Mass (g) 10 10 10 20.0 Theoretical Sample 0.0 8.6 4.2
 12.5 Mass (g) Biodegraded Mass 0.60 0.48 4.29 2.84 (g) Percent Biode- -1.4 87.4 18.0 graded (%)
 * Adjusted Percent -1.6 100.0 20.6 Biodegraded (%)

[0347] The biodegradation is shown in FIG. 1 versus the control.

EXAMPLE 2

[0348] A bi-component fibre with polyethylene terephthalate (PET) as core (thermoplastic polymer A) and polypropylene (PP) as shell (sheath) (thermoplastic polymer B) is spun from a polyethylene terephthalate (PET) resin and a polypropylene (PP) resin having the following properties:

[0349] The melt extrusion is done by an extruder having one or more screws at a temperature of 270° C. for PET and by another extruder having one or more screws at a temperature at a temperature of 250° C. for PP.

[0350] Additive A is added to the PET at the extruder feed-throat at a level of 2 wt.-% masterbatch dosage. This masterbatch consists of a PET polyester as carrier and the additive, which comprises an aliphatic polyester and CaCO.sub.3.

[0351] Additive B is added to the PP at the extruder feed-throat at a level of 2 wt.-% masterbatch dosage. This masterbatch consists of PP as carrier and the additive, which comprises transition metal compounds and unsaturated carboxylic acids.

[0352] The fibre quench occurs by crossflow and air temperature of 20° C.; fibre drawdown speed is 1000 m/min. Spun fiber fineness is 5.4 dtex.

[0353] The fibre drawing is done by can be single or duo-stage drawing with draw ratio up to 4 and the final dtex is 2.5 dtex. Heat setting is done at 110-130° C.

[0354] The fibre produced is cut into staple fibre having a length of 38 mm and a nonwoven is produced by thermo-bonding.

[0355] A nonwoven thus produced is kept as control in a sealed, evacuated bag and another nonwoven thus produced is tested over a one year period (365 days) at 60° C. and 60% relative humidity.

[0356] The degradation is shown in FIG. 2 *a-e* versus the control. The degradation of the PP sheath becomes clearly visible. FIG. 2*e* illustrates the degradation of the PET core as the shape of the fraction changes clearly from the mushroom-shape, which is typical for PET, to a shape indicating that the material has become brittle. In this test the fibers have been axially stressed in a reproducible way (defined speed) by a mechanical testing machine.

[0357] The core of this bico fiber has the same material composition (polymer and additive) as the fiber described in Example 1, where degradation has been proven according to ASTM D5511.

EXAMPLE 3

[0358] A bi-component fibre with polyethylene terephthalate (PET) as core (thermoplastic polymer A) and co-polyethylene terephthalate (coPET) as shell (sheath) (thermoplastic polymer B) is spun from a polyethylene terephthalate (PET) resin and a co-polyethylene terephthalate (coPET) resin having the following properties:

[0359] The melt extrusion is done by an extruder having one or more screws at a temperature of 290° C. for PET and by another extruder having one or more screws at a temperature at a temperature of 280° C. for coPET.

[0360] Additive A is added to the PET at the extruder feed-throat at a level of 2 wt.-% masterbatch dosage. This masterbatch consists of a PET polyester as carrier and the additive, which comprises an aliphatic polyester and CaCO.sub.3.

[0361] Additive B is added to the coPET at the extruder feed-throat at a level of 2 wt.-% masterbatch dosage. This additive B is identical to additive A.

[0362] The fibre quench occurs by crossflow and air temperature of 35° C.; fibre drawdown speed is 1200 m/min. Spun fiber fineness is 5.4 dtex.

[0363] The fibre drawing is done by single or duo-stage drawing with draw ratio up to 4.5 and the final dtex is 2.5 dtex. Heat setting is done at 80° C.

[0364] The fibre produced is cut into staple fibre having a length of 38 mm and a nonwoven is produced by thermo-bonding.

[0365] The resulting fiber meets all requirements imposed. The core of this bico fiber has the same material composition (polymer and additive) as the fiber described in Example 1, where degradation has been proven according to ASTM D5511. The sheath differs in that the melting point of the copolyester is lower than the polyester of the core, which renders possible to use the fiber for thermobonded nonwovens.

Claims

1. A multi-component polymer fibre containing (i) at least one component A and at least one component B, (ii) the component A comprising a thermoplastic polymer A, (iii) the component B comprising a thermoplastic polymer B, characterized in that (iv) the component A additionally has at least one additive A which increases the biological degradability of the multi-component fibre and the component B does not have an additive B which increases the biological degradability of the multi-component fibre, or (v) the component B additionally has at least one additive B which increases the biological degradability of the multi-component fibre and the component A does not have an additive A which increases the biological degradability of the multi-component fibre, or (vi) the component A additionally has at least one additive A and the component B additionally has at least one additive B which together increase the biological degradability of the multi-component fibre, with the proviso that when (i) the thermoplastic polymer A and the thermoplastic polymer B are identical, the additives A and B are different, or (ii) when the additives A and B are identical, thermoplastic polymer A and thermoplastic polymer B are different, said multi-component polymer fibre has an increased biological degradability compared with a multi-component fibre without the additives A and/or B, and the biological degradability is determined in accordance with at least one method selected from the group: (i) ASTM D5338-15 (2021) Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials Under Controlled Composting Conditions, Incorporating Thermophilic Temperatures (DOI: 10.1520/D5338-15R21) ASTM International, West Conshohocken, PA, 2015, www.astm.org), (ii) ASTM D6400-12 (Standard Specification for Labeling of Plastics Designed to be Aerobically Composted in Municipal or Industrial Facilities) (DOI: 10.1520/D6400-12), (iii) ASTM D5511 (ASTM D5511-11 Standard Test Method for Determining Anaerobic Biodegradation of Plastic Materials Under High-Solids Anaerobic-digestion Conditions (DOI: 10.1520/D5511-11) and ASTM D5511-18 Standard Test Method for Determining Anaerobic Biodegradation of Plastic Materials Under High-Solids Anaerobic-digestion Conditions; (DOI: 10.1520/D5511-18)), (iv) ASTM D6691 (ASTM D6691-09 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in the Marine Environment by a Defined Microbial Consortium or Natural Sea Water Inoculum) (DOI: 10.1520/D6691-09) and ASTM D6691-17, Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in the Marine Environment by a Defined Microbial Consortium or Natural Sea Water Inoculum (DOI: 10.1520/D6691-17)), (v) ASTM D5210-92 (Anaerobic Degradation in the Presence of Sewage Sludge) (DOI: 10.1520/D5210-92), (vi) PAS 9017:2020 (Plastics—Biodegradation of polyolefins in an open-air terrestrial environment—Specification), ISBN 978 0 539 17478 6; 2021 Oct. 31, (vii) ASTM D5988 (ASTM D5988-12 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in Soil) (DOI: 10.1520/D5988-12), ASTM D5988-18 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in Soil (DOI: 10.1520/D5988-18), ASTM D5988-03 Standard Test Method for Determining Aerobic Biodegradation in Soil of Plastic Materials or Residual Plastic Materials After Composting (DOI: 10.1520/D5988-03)), (viii) EN 13432:2000-12

Packaging—Requirements for packaging recoverable through composting and biodegradation—Test scheme and evaluation criteria for the final acceptance of packaging; German version EN 13432:2000 (DOI: 10.31030/9010637), (ix) ISO 14855-1:2013-04 (DOI: 10.31030/1939267) and ISO 14855-2:2018-07 (ICS 83.080.01) Determination of the ultimate aerobic biodegradability of plastic materials under controlled composting conditions (Method by analysis of evolved carbon dioxide), (x) EN 14995:2007-03—Plastics—Evaluation of compostability (DOI: 10.31030/9730527) or (xii) ISO 17088:2021-04 (Specifications for compostable plastics) (ICS 83.080.01).

2. The multi-component polymer fibre as claimed in claim 1, characterized in that the thermoplastic polymer A and/or the thermoplastic polymer B is/are selected from the group: (i) acrylonitrile-ethylene-propylene-(diene)-styrene copolymer, acrylonitrile-methacrylate copolymer, acrylonitrile-methylmethacrylate copolymer, chlorinated acrylonitrile, polyethylene-styrene copolymer, acrylonitrile-butadiene-styrene copolymer, acrylonitrile-ethylene-propylene-styrene copolymer, cellulose acetobutyrate, cellulose acetopropionate, hydrated cellulose, carboxymethylcellulose, cellulose nitrate, cellulose propionate, cellulose triacetate, polyvinyl chloride, ethylene-acrylic acid copolymer, ethylene-butylacrylate copolymer, ethylene-chlorotrifluoroethylene copolymer, ethylene-ethylacrylate copolymer, ethylene-methacrylate copolymer, ethylene-methacrylic acid copolymer, ethylene-tetrafluoroethylene copolymer, ethylene-vinyl alcohol copolymer, ethylene-butene copolymer, ethylcellulose, polystyrene, polyfluoroethylene-propylene, methylmethacrylate-acrylonitrile-butadiene styrene copolymer, methylmethacrylate-butadiene-styrene copolymer, methylcellulose, polyamide 11, polyamide 12, polyamide 46, polyamide 6, polyamide 6-3-T, polyamide 6-terephthalic acid copolymer, polyamide 66, polyamide 69, polyamide 610, polyamide 612, polyamide 61, polyamide MXD 6, polyamide PDA-T, polyamide, polyarylether, polyaryletherketone, polyamideimide, polyarylamide, polyamino-bis-maleimide, polyarylate, polybutene-1, polybutylacrylate, polybenzimidazole, poly-bis-maleimide, polyoxadiazobenzimidazole, polybutylterephthalate, polycarbonate, polychlorotrifluoroethylene, polyethylene, polyestercarbonate, polyaryletherketone, polyetheretherketone, polyetherimide, polyetherketone, polyethylene oxide, polyarylether sulphone, polyethylene terephthalate, polyimide, polyisobutylene, polyisocyanurate, polyimide sulphone, polymethacrylimide, polymethacrylate, poly-4-methylpentene, polyacetal, polypropylene, polyphenyl oxide, polypropylene oxide, polyphenylene sulphide, polyphenylene sulphone, polystyrene, polysulphone, polytetrafluoroethylene, polyurethane, polyvinyl acetate, polyvinyl alcohol, polyvinylbutyral, polyvinyl chloride, polyvinylidene chloride, polyvinylidene fluoride, polyvinyl fluoride, polyvinyl methyl ether, polyvinylpyrrolidone, styrene-butadiene copolymer, styrene-isoprene copolymer, styrene-maleic acid anhydride copolymer, styrene-maleic acid anhydride-butadiene copolymer, styrene methyl methacrylate copolymer, styrene-methyl styrene copolymer, styrene-acrylonitrile copolymer, vinyl chloride-ethylene copolymer, vinyl chloride-methacrylate copolymer, vinyl chloride-maleic acid anhydride copolymer, vinyl chloride-maleimide copolymer, vinyl chloride-methylmethacrylate copolymer, vinyl chloride-octyl acrylate copolymer, vinyl chloride-vinyl acetate copolymer, vinyl chloride-vinylidene chloride copolymer, vinyl chloride-vinylidene chloride-acrylonitrile copolymer and/or (ii) synthetic biopolymer.

3. The multi-component polymer fibre as claimed in claim 2, characterized in that the synthetic biopolymers may be one or more aliphatic, araliphatic polyesters or copolyesters which are produced from polyols, and aliphatic and/or aromatic dicarboxylic acids or their derivatives (anhydrides, esters) by polycondensation, wherein the polyols may be substituted or unsubstituted, and the polyols may be linear or branched polyols.

4. The multi-component polymer fibre as claimed in claim 3, characterized in that (i) the polyols contain 2 to 8 carbon atoms, (ii) the aliphatic dicarboxylic acids comprise substituted or unsubstituted, linear or branched, non-aromatic dicarboxylic acids selected from the group formed by aliphatic dicarboxylic acids containing 2 to 12 carbon atoms and cycloaliphatic dicarboxylic

acids containing 5 to 10 carbon atoms, wherein the cycloaliphatic dicarboxylic acids may also contain heteroatoms in the ring, (iii) the aromatic dicarboxylic acids comprise substituted or unsubstituted, aromatic dicarboxylic acids selected from the group formed by aromatic dicarboxylic acids containing 6 to 12 carbon atoms, wherein these carboxylic acids may also comprise heteroatoms in the aromatic ring and/or in the substituents, (iv) the substituted aromatic dicarboxylic acids contain 1 to 4 substituents selected from halogens, C6-C10 aryl and C1-C4 alkoxy.

5. The multi-component polymer fibre as claimed in claim 1, characterized in that the synthetic biopolymer is selected from the group formed by aliphatic polyesters with repeat units of at least 4 carbon atoms, for example polyhydroxyalkanoates such as polyhydroxyvalerate and polyhydroxybutyrate-hydroxyvalerate copolymer, polycaprolactone, furan dicarboxylic acid, and succinate-based aliphatic polymers (for example polybutylene succinate, polybutylene succinate adipate and polyethylene succinate). Special examples may be selected from polyethylene oxalate, polyethylene malonate, polyethylene succinate, polypropylene oxalate, polypropylene malonate, polypropylene succinate, polybutylene oxalate, polybutylene malonate, polybutylene succinate and blends and copolymers of these compounds.

6. The multi-component polymer fibre as claimed in claim 1, characterized in that the synthetic biopolymer is an aliphatic polyester comprising repeat units of lactic acid (PLA), hydroxy fatty acid (PHF) (also known as polyhydroxyalkanoate PHA), in particular hydroxybutanoic acid (PHB) and succinate-based aliphatic polymers, for example polybutylene succinate, polybutylene succinate adipate and polyethylene succinate).

7. The multi-component polymer fibre as claimed in claim 1, characterized in that the thermoplastic polymer A and/or B has a glass transition temperature in the range -125°C. to 200°C. , in particular in the range -125°C. to 100°C.

8. The multi-component polymer fibre as claimed in claim 1, characterized in that the thermoplastic polymer A and/or B has a melting temperature in the range 120°C. to 285°C. , in particular in the range 150°C. to 270°C. , particularly preferably in the range 175°C. to 270°C.

9. The multi-component polymer fibre as claimed in claim 1, characterized in that the thermoplastic polymer A and/or B is/are selected from the group formed by polylactic acids (PLA) as well as their copolymers, polyhydroxy fatty acid esters (PHF) as well as their copolymers, as well as blends of said polymers.

10. The multi-component polymer fibre as claimed in claim 1, characterized in that at least the thermoplastic polymer A and/or the thermoplastic polymer B is/are selected from the group formed by melt-spinnable synthetic biopolymers, wherein polycondensates and polymerisates from bio-based starting materials are particularly preferred.

11. The multi-component polymer fibre as claimed in claim 1, characterized in that the multi-component polymer fibre is a bi-component fibre in which the component A forms the core and the component B forms the shell and the melting point of the thermoplastic polymer in component A is at least 5°C. , preferably at least 10°C. , higher than the melting point of the thermoplastic polymer in component B.

12. The multi-component polymer fibre as claimed in claim 1, characterized in that the fibre has (i) at least one additive A in the component A or (ii) at least one additive B in the component B or (iii) at least one additive A in the component A and at least one additive B in the component B, with the proviso that the additive A and the additive B are different or insofar as at least one additive A is present in the component A and at least one additive B is present in the component B, the additive A and the additive B may also be identical, when the thermoplastic polymer A and thermoplastic polymer B are different.

13. The multi-component polymer fibre as claimed in claim 1, characterized in that the additives A and B are selected from the group: (i) basic alkali and/or alkaline earth compounds ($\text{pH} > 7$ dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly

preferably CaCO₃, and alkaline additives, particularly preferably CaO, (ii) aliphatic polyesters, (iii) fatty acid ester, preferably C1-C40-alkyl stearate, more preferred C2-C20-alkyl stearate, most preferred ethyl stearate (iv) sugars, in particular monosaccharides, disaccharides and oligosaccharides, (v) catalysts for transesterification, in particular under basic conditions, (vi) metal compounds, in particular transition metal compounds, as well as their salts, (vii) unsaturated carboxylic acids or their anhydrides/esters/amides, (viii) synthetic rubber, natural rubber, (ix) carbohydrates, in particular starch and/or cellulose, as well as mixtures of the aforementioned substances.

14. The multi-component polymer fibre as claimed in claim 1, characterized in that the additive A has a proportion of the component A which is preferably between 0.005% by weight and 20% by weight, particularly preferably between 0.01% by weight and 5% by weight, with respect to the total weight of the component A and the additive B has a proportion of the component B which is preferably between 0.005% by weight and 20% by weight, particularly preferably between 0.01% by weight and 5% by weight, with respect to the total weight of the component B.

15. The multi-component polymer fibre as claimed in claim 1, characterized in that the fibre is a continuous fibre, preferably a staple fibre, or is a continuous filament, and is preferably a bi-component fibre.

16. A bi-component fibre with a core/shell structure, wherein (i) the component A forms the core and the component B forms the shell of the fibre, (ii) the component A in the core comprises thermoplastic polymer A, (iii) the component B comprises a thermoplastic polymer B, (iv) the melting point of the thermoplastic polymer in the component A in the core is at least 5° C. higher than the melting point of the thermoplastic polymer in the component B in the shell, and preferably the melting point is at least 10° C. higher, characterized in that (v) the component A has a higher biological degradability than the component B; preferably, the component A has at least one additive A, or (vi) the component B has a higher biological degradability than the component A; preferably, the component B has at least one additive B, said biological degradability is determined in accordance with at least one method selected from the group: (i) ASTM D5338-15 (2021) Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials Under Controlled Composting Conditions, Incorporating Thermophilic Temperatures (DOI: 10.1520/D5338-15R21) ASTM International, West Conshohocken, PA, 2015, www.astm.org), (ii) ASTM D6400-12 (Standard Specification for Labeling of Plastics Designed to be Aerobically Composted in Municipal or Industrial Facilities) (DOI: 10.1520/D6400-12), (iii) ASTM D5511 (ASTM D5511-11 Standard Test Method for Determining Anaerobic Biodegradation of Plastic Materials Under High-Solids Anaerobic-digestion Conditions (DOI: 10.1520/D5511-11) and ASTM D5511-18 Standard Test Method for Determining Anaerobic Biodegradation of Plastic Materials Under High-Solids Anaerobic-digestion Conditions; (DOI: 10.1520/D5511-18)), (iv) ASTM D6691 (ASTM D6691-09 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in the Marine Environment by a Defined Microbial Consortium or Natural Sea Water Inoculum) (DOI: 10.1520/D6691-09) and ASTM D6691-17, Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in the Marine Environment by a Defined Microbial Consortium or Natural Sea Water Inoculum (DOI: 10.1520/D6691-17)), (v) ASTM D5210-92 (Anaerobic Degradation in the Presence of Sewage Sludge) (DOI: 10.1520/D5210-92), (vi) PAS 9017:2020 (Plastics—Biodegradation of polyolefins in an open-air terrestrial environment—Specification), ISBN 978 0 539 17478 6; 2021 Oct. 31, (vii) ASTM D5988 (ASTM D5988-12 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in Soil) (DOI: 10.1520/D5988-12), ASTM D5988-18 Standard Test Method for Determining Aerobic Biodegradation of Plastic Materials in Soil (DOI: 10.1520/D5988-18), ASTM D5988-03 Standard Test Method for Determining Aerobic Biodegradation in Soil of Plastic Materials or Residual Plastic Materials After Composting (DOI: 10.1520/D5988-03)), (viii) EN 13432:2000-12 Packaging—Requirements for packaging recoverable through composting and biodegradation—

Test scheme and evaluation criteria for the final acceptance of packaging; German version EN 13432:2000 (DOI: 10.31030/9010637), (ix) ISO 14855-1:2013-04 (DOI: 10.31030/1939267) and ISO 14855-2:2018-07 (ICS 83.080.01) Determination of the ultimate aerobic biodegradability of plastic materials under controlled composting conditions (Method by analysis of evolved carbon dioxide), (x) EN 14995:2007-03—Plastics—Evaluation of compostability (DOI: 10.31030/9730527) or (xi) ISO 17088:2021-04 (Specifications for compostable plastics) (ICS 83.080.01).

17. (canceled)

18. (canceled)

19. The bi-component fibre as claimed in claim 16, characterized in that the additive A and/or additive B is selected from the group formed by (i) basic alkali and/or alkaline earth compounds (pH>7 dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly preferably CaCO_3 , and alkaline additives, particularly preferably CaO, (ii) aliphatic polyesters, (iii) fatty acid ester, preferably C1-C40-alkyl stearate, more preferred C2-C20-alkyl stearate, most preferred ethyl stearate, (iv) sugars, in particular mono-saccharides, di-saccharides and oligo-saccharides, (v) catalysts for transesterifications, in particular under basic conditions, (vi) carbohydrates, in particular starch and/or cellulose, as well as mixtures thereof.

20. The bi-component fibre as claimed in claim 16, characterized in that the thermoplastic polymer A and/or the thermoplastic polymer B comprises at least one polyester, with the proviso that the polyester is an aliphatic polyester or copolyester in the case in which the additive A and/or B is an aliphatic polyester.

21. The bi-component fibre as claimed in claim 16, characterized in that the additive A and/or additive B is selected from the group formed by A) basic alkali and/or alkaline earth compounds (pH>7 dissolved in water), in particular carbonates, hydrogen carbonates, sulphates, particularly preferably CaCO_3 , and alkaline additives, particularly preferably CaO in combination with catalysts for transesterifications, in particular under basic conditions; B) sugars, in particular mono-saccharides, di-saccharides and oligo-saccharides, in combination with carbohydrates, in particular starch and/or cellulose, as well as mixtures thereof; C) aliphatic polyesters in combination with sugars, in particular mono-saccharides, di-saccharides and oligo-saccharides, or carbohydrates, in particular starch and/or cellulose, D) fatty acid ester, preferably C1-C40-alkyl stearate, more preferred C2-C20-alkyl stearate, most preferred ethyl stearate, as well as mixtures thereof.

22. Use of the multi-component polymer fibre as claimed in claim 1, for the production of a textile fabric.

23. A textile fabric containing the multi-component polymer fibres as claimed in claim 1.

24. A textile fabric as claimed in claim 21, characterized in that the textile fabric is a nonwoven, in particular a wet laid nonwoven or a dry laid nonwoven, preferably based on staple fibres, wherein the nonwoven is preferably consolidated by thermobonding.

25. The textile fabric as claimed in claim 21, characterized in that the textile fabric, in particular the nonwoven, has a basis weight between 10 and 500 g/m², preferably 25 to 450 g/m², in particular 30 to 300 g/m².
